IMU-Free Body-Frame State Estimation with Sparse Scene Flow for Quadcopters

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Abstract

We present a vision-only state estimation system for X-configuration quadcopters equipped with a canonical stereo camera pair and no inertial sensors. The system operates entirely in the body frame, requiring only synchronised stereo images and motor thrust commands as inputs. A continuous-discrete extended Kalman filter on a composite manifold state $\langle SE(3), \mathbb{R}^3, \ldots \rangle$ maintains estimates of body-frame pose, velocity, angular velocity, gravity, and disturbances, using stationary scene points as implicit inertial references. Feature points are detected (FAST, Shi-Tomasi), tracked temporally (SSD, Lucas-Kanade) and matched across cameras (NCC), with prediction-assisted search regions constructed from filter-derived pose and point uncertainty. Chi-squared gating on the normalised innovation classifies points as stationary or moving; only stationary points enter the filter. Beyond the state estimate, the system produces a sparse 3D point cloud in which each point carries a position, velocity, and joint covariance. These are obtained through a 4-view (i.e. two stereo pairs for two time stamps) full bundle adjustment that jointly estimates point position and velocity, fusing stereo disparity and temporal parallax across all visibility combinations (SS, SM, MS, MM), using the filter-derived relative pose as a prior. Feature points in the EKF do not enter the solver; their measurement information is reflected through the pose prior. Point cloud density is spatially adaptive: an external focus point directs allocation, producing dense coverage in the region of attention and sparse peripheral coverage from the feature points. The output — a body-frame state estimate, a calibrated pose change, and a sparse scene flow — is intended as a measurement source for a downstream world model anchored in the current body frame, without dependence on GPS, IMU, or any world-frame infrastructure, though the architecture accommodates their future integration.

Contents

1 Introduction

This document describes the design of a vision-only state and scene flow estimation system. The system takes synchronised stereo images and motor thrust commands as input and produces three outputs: (1) the drone's body-frame state (pose, velocity, angular velocity, gravity, disturbance) with covariance, (2) the frame-to-frame pose change with covariance, and (3) a sparse 3D point cloud with per-point velocity and uncertainty.

The architecture has two coupled estimators. The first is a continuous-discrete extended Kalman filter on a composite manifold state $\langle SE(3), \mathbb{R}^3, \ldots \rangle$ (Section 2.2). The filter maintains stationary scene points as inertial references: their body-frame positions evolve deterministically under the known kinematics (Section 2.5), and pixel measurements correct the state through covariance cross-correlations (Section 2.6). A gravity magnitude pseudo-measurement prevents drift in the gravity estimate.

The second estimator is a two frame pair nonlinear MAP solver (Section 2.8.7) that jointly estimates point positions and velocities with a pose change, using the filter-derived pose as a Lie algebra prior (Section 2.8.6) and stereo triangulated point estimates as priors on the points (Section 2.9). Feature points in the EKF do not enter the solver; their information enters through the pose prior, avoiding double-counting. The solver handles all four stereo/mono visibility combinations (SS, SM, MS, MM), with mono-mono points constrained to the observable perpendicular velocity component (Section 2.8.7).

Points are managed in three sets: feature points ($\mathcal{F}$) in the EKF for state estimation, pre-admission candidates ($\mathcal{F}_{\text{pre}}$) awaiting their first solve and stationarity test, and interest points ($\mathcal{I}$) steered by an external focus for task-directed dense coverage. All candidate and interest points enter the joint solver; the output point cloud is uniform across roles (Section 3).

The system operates without IMU, GPS, or any world-frame infrastructure. The body-frame formulation means the output is naturally suited to a world model anchored in the current body frame B_k, propagated between frames using the solver's pose change $\Delta\hat{T}_{\text{solver}}$ rather than an accumulating absolute pose.

1.0.1 Rotor Configuration

Viewed from above, the rotors are numbered and spin as follows:

Rotor	Position	Spin	Body-frame location
1	Front-right	CW	$+e_1,\ +e_2$
2	Rear-right	CCW	$-e_1,\ +e_2$
3	Rear-left	CW	$-e_1,\ -e_2$
4	Front-left	CCW	$+e_1,\ -e_2$

Adjacent rotors spin in opposite directions to cancel net reactive torque in steady flight. CW rotors $(1, 3)$ produce positive yaw torque (about $+e_3$); CCW rotors $(2, 4)$ produce negative yaw torque. All rotors produce thrust along $-e_3$ (upward in FRD, opposing gravity).

1.0.2 Camera Configuration

Two hardware-synchronised cameras with global shutter are rigidly mounted facing forward (along $+e_1$), arranged in a canonical stereo configuration: horizontally separated along e_2 with parallel optical axes. The left camera is at $-e_2$ and the right camera at $+e_2$ relative to the body centre, giving a baseline $b = \|\mathbf{t}_R - \mathbf{t}_L\|$.

Both cameras share the same orientation: their optical axes are aligned with the body e_1 axis, image x-axis with e_2, and image y-axis with e_3. The stereo overlap region (where both fields of view intersect) is centred on the forward direction. The left and right fringe zones are visible to only one camera.

The cameras are fixed-focus with deep depth of field. Images are delivered as synchronised $w \times h$ frames at rate λ Hz. Fish-eye lenses are preferred for wider field of view; HDR or dual-exposure capability is beneficial for varying lighting conditions.

1.1 Priors

All quantities that must be known or calibrated before the algorithm runs. These are constant throughout operation. Runtime values, units, sources, and mathematical types (manifolds, shapes) are maintained in calibration.yaml. Groups, with cross-references for how each is used:

- **Coordinate frames.** Body B (FRD), cameras C^L, C^R (RealSense), Vicon marker M. Used throughout.
- **Camera parameters.** Intrinsics, distortion, body-to-camera extrinsics in $SE(3)$, frame rate, image resolution. Stereo baseline b derived. See §2.3.
- **Drone parameters.** Mass, inertia, arm length, per-rotor thrust and torque coefficients, drag, gravity. Determine mixing matrix and dynamics. See §2.4.
- **Noise parameters.** Process noise spectral densities for actuators, torques, gravity, and disturbance; pixel measurement noise; gravity pseudo-measurement noise. See §2.5, §2.6.
- **Initial covariance.** Diagonal of P_0 for each state component, reflecting the assumption that the drone starts stationary, level, at the body-frame origin. See §2.5.

1.2 Inputs

All objects the algorithm receives at the start of time step t_k. The control input and focus are indexed $k-1$ because they reflect commands applied during the interval

$[t_{k-1},\, t_k)$:

$$\begin{aligned} L_k,\ R_k &\in \mathbb{R}^{3\times h\times w} && \text{Stereo pair with timestamp } t_k && (1)\\ \mathbf{u}_{k-1} &= (T_1,\, T_2,\, T_3,\, T_4)^\top \in \mathbb{R}^4 && \text{Rotor thrusts during } [t_{k-1},\, t_k) \text{ (§2.5)} && (2)\\ \mathbf{F}_{k-1} &\in \mathbb{R}^3 && \text{Focus point in } B_{k-1} \text{ (§2.7)} && (3)\\ \sigma_{F,k-1} &\in \mathbb{R}_{>0} && \text{Focus spread (§2.7)} && (4) \end{aligned}$$

1.3 Outputs

All objects the algorithm produces at the end of time step t_k.

1.3.1 Core State (body frame)

The EKF state with feature point rows and columns stripped (§3):

$$\begin{aligned} \hat{T}_{B_k,B_0} &\in SE(3) && \text{Body pose (drifts over time)} && (5)\\ \hat{\mathbf{v}} &\in \mathbb{R}^3 && \text{Linear velocity in } B_k && (6)\\ \hat{\boldsymbol{\omega}} &\in \mathbb{R}^3 && \text{Angular velocity in } B_k && (7)\\ \hat{\mathbf{g}}^B &\in \mathbb{R}^3 && \text{Gravity in } B_k\ (\|\hat{\mathbf{g}}^B\| \approx g) && (8)\\ \hat{\mathbf{d}}^B &\in \mathbb{R}^3 && \text{Wind/disturbance in } B_k && (9)\\ \mathbf{P}^{\text{core}} &\in \mathbb{R}^{18\times 18} && \text{Core state covariance} && (10) \end{aligned}$$

The covariance is 18×18. The pose $\hat{T}_{B_k,B_0}$ accumulates drift and is suitable for coarse waypoint navigation; for local geometry, use $\Delta\hat{T}$ below.

1.3.2 Pose Change

From the joint solver (§2.8.7):

$$\begin{aligned} \Delta\hat{T}_{\text{solver}} &= T_{B_k,B_{k-1}} \in SE(3) && \text{Frame-to-frame pose change} && (11)\\ \boldsymbol{\Sigma}^{\text{solver}}_{\Delta\xi} &\in \mathbb{R}^{6\times 6} && \text{Pose change covariance (Lie algebra)} && (12) \end{aligned}$$

This is the recommended transform for propagating a downstream world model anchored in the current body frame (§3).

1.3.3 Point Cloud (body frame B_k)

$$\mathcal{C}_k = \{(\hat{\mathbf{p}}_i,\ \hat{\mathbf{v}}^{\mathbf{P}}_i,\ \boldsymbol{\Sigma}_i,\ \texttt{role}_i,\ \texttt{stage}_i)\}_{i=1}^{N} \qquad (13)$$

where for each point i:

$$\begin{aligned} \hat{\mathbf{p}}_i &\in \mathbb{R}^3 && \text{Position in } B_k && (14)\\ \hat{\mathbf{v}}^{\mathbf{P}}_i &\in \mathbb{R}^3 && \text{Velocity in } B_k \text{ (stage 2 only)} && (15)\\ \boldsymbol{\Sigma}_i &\in \mathbb{R}^{d_i\times d_i} && \text{Covariance (dimension depends on source)} && (16)\\ \texttt{role}_i &\in \{\mathcal{F},\, \mathcal{F}_{\text{pre}},\, \mathcal{I}\} && \text{Point role} && (17)\\ \texttt{stage}_i &\in \{1,\, 2\} && \text{Single-frame or two-frame} && (18) \end{aligned}$$

The content depends on the point's source:

Stage 2 ($\mathcal{F}_{\text{pre}}$, $\mathcal{I}$). Position, velocity, and joint covariance from the joint solver, transformed to B_k (§2.8.7):

$$\boldsymbol{\Sigma}_i^{B_k} = \mathbf{T}_{\Delta,i}\,\boldsymbol{\Sigma}_{s_i}\,\mathbf{T}_{\Delta,i}^\top \tag{19}$$

For SS/SM/MS: $\boldsymbol{\Sigma}_i \in \mathbb{R}^{6\times 6}$ (full position-velocity). For MM: $\boldsymbol{\Sigma}_i \in \mathbb{R}^{4\times 4}$ (position and perpendicular velocity $v_\perp$); the velocity field is $\hat{\mathbf{v}}_i^{\mathbf{P}} = \hat{v}_{\perp,i}\,\Delta\hat{\mathbf{R}}\,\mathbf{d}_\perp$ with the along-epipolar component unobservable and set to zero (§2.8.7).

Stage 1 stereo ($\mathcal{F}_{\text{pre}}$, $\mathcal{I}$). Position and covariance from stereo disparity (§2.9): $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}_{p,i} \in \mathbb{R}^{3\times 3}$. Velocity not yet available.

Feature points ($\mathcal{F}$). Position and marginal covariance extracted from the EKF state and covariance: $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}_{p,i}^{\text{EKF}} \in \mathbb{R}^{3\times 3}$. Velocity not returned (the EKF does not maintain per-point velocity; feature points are assumed stationary, and this assumption is repeatedly tested).

2 Theory

2.1 Lie Algebra on $SE(3)$

We consider the matrix Lie group $T \in SE(3)$, $T = \begin{pmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{pmatrix}$, with Lie algebra elements $\boldsymbol{\xi}^\wedge \in \mathfrak{se}(3)$ and coordinate vectors $\boldsymbol{\xi} \in \mathbb{R}^6$ (Solà et al., 2021).
A coordinate vector and its Lie algebra element are:

$$\boldsymbol{\rho} = \text{Translation}, \quad \boldsymbol{\phi} = \text{Rotation}$$
$$\boldsymbol{\xi} = \begin{pmatrix} \boldsymbol{\rho} \\ \boldsymbol{\phi} \end{pmatrix} = \begin{pmatrix} \rho_1 & \rho_2 & \rho_3 & \phi_1 & \phi_2 & \phi_3 \end{pmatrix}^\top$$
$$\boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\phi}]_\times & \boldsymbol{\rho} \\ \mathbf{0}^\top & 0 \end{bmatrix} \in \mathbb{R}^{4\times 4}, \qquad (\boldsymbol{\xi}^\wedge)^\vee = \boldsymbol{\xi}$$

The rotation $\boldsymbol{\phi}$ can be represented in angle-axis form $\boldsymbol{\phi} = \theta\mathbf{u}$. A coordinate vector is related to a pose and vice versa by (where exp is the matrix exponential and log its inverse):

$$T_{\boldsymbol{\xi}} = \mathrm{Exp}(\boldsymbol{\xi}) = \exp(\boldsymbol{\xi}^\wedge), \qquad \boldsymbol{\xi}_T = \mathrm{Log}(T) = \log(T)^\vee$$

$$\begin{aligned}
\mathbf{R} = \mathrm{Exp}_{so(3)}(\boldsymbol{\phi}) &:= \mathbf{I} + \sin\theta[\mathbf{u}]_\times + (1-\cos\theta)[\mathbf{u}]_\times^2 \\
\boldsymbol{\phi} = \mathrm{Log}_{so(3)}(\mathbf{R}) := \frac{\theta}{2\sin\theta}(\mathbf{R}-\mathbf{R}^\top)^\vee, &\qquad \theta = \arccos\left(\frac{\mathrm{tr}(\mathbf{R})-1}{2}\right) \\
\mathbf{V}(\boldsymbol{\phi}) &= \mathbf{I} + \frac{1-\cos\theta}{\theta}[\mathbf{u}]_\times + \frac{\theta-\sin\theta}{\theta}[\mathbf{u}]_\times^2 \\
\mathbf{V}^{-1}(\boldsymbol{\phi}) &= \mathbf{I} - \frac{\theta}{2}[\mathbf{u}]_\times + \left(1 - \frac{\theta\sin\theta}{2(1-\cos\theta)}\right)[\mathbf{u}]_\times^2
\end{aligned}$$

$$\begin{aligned}
\mathrm{Exp}(\boldsymbol{\xi}) &:= \begin{bmatrix} \mathrm{Exp}_{so(3)}(\boldsymbol{\phi}) & \mathbf{V}(\boldsymbol{\phi})\boldsymbol{\rho} \\ \mathbf{0}^\top & 1 \end{bmatrix} \\
\mathrm{Log}(T) &:= \begin{bmatrix} \mathbf{V}^{-1}(\boldsymbol{\phi})\mathbf{t} \\ \boldsymbol{\phi} \end{bmatrix}
\end{aligned}$$

Addition and subtraction on this Lie group (right perturbation convention):

$$\begin{aligned}
T \oplus \boldsymbol{\xi} &= T \cdot \mathrm{Exp}(\boldsymbol{\xi}) \in SE(3) \\
T_1 \ominus T_2 &= \mathrm{Log}(T_2^{-1} \cdot T_1) \in \mathbb{R}^6
\end{aligned}$$

Adjoint of $SE(3)$. The adjoint matrix Ad_T linearly maps tangent vectors between reference frames ((Solà et al., 2021), Example 6, Appendix D Eq. 175):

$$\mathrm{Ad}_T = \begin{pmatrix} \mathbf{R} & [\mathbf{t}]_\times\mathbf{R} \\ \mathbf{0} & \mathbf{R} \end{pmatrix} \in \mathbb{R}^{6\times6} \tag{20}$$

satisfying $(\,T\,\boldsymbol{\xi}^\wedge\,T^{-1}\,)^\vee = \mathrm{Ad}_T\,\boldsymbol{\xi}$.

2.1.1 Composite Manifold State

The estimation state contains both Lie group and Euclidean components. Following ((Solà et al., 2021) , Section IV, Eqs. 84–90), we define a composite manifold $\mathcal{M} = \langle SE(3), \mathbb{R}^3, \mathbb{R}^3, \ldots\rangle$ with state:

$$\mathcal{X} = \langle\, T,\ \mathbf{v},\ \boldsymbol{\omega},\ \mathbf{g}^B,\ \mathbf{d}^B,\ \mathbf{p}_1^B,\ \ldots,\ \mathbf{p}_{N_f}^B\,\rangle \tag{21}$$

where $T \in SE(3)$ is the body pose and all other components are in $\mathbb{R}^3$.

Composite plus and minus. The composite operators $\hat{\oplus}$ and $\hat{\ominus}$ act on each block according to its own manifold ((Solà et al., 2021), Eqs. 86–87):

$$\mathcal{X} \hat{\oplus} \delta\mathbf{x} = \langle\, T \oplus \delta\boldsymbol{\xi},\ \mathbf{v}+\delta\mathbf{v},\ \boldsymbol{\omega}+\delta\boldsymbol{\omega},\ \mathbf{g}^B+\delta\mathbf{g},\ \mathbf{d}^B+\delta\mathbf{d}^B,\ \mathbf{p}_1^B+\delta\mathbf{p}_1,\ \ldots\rangle \tag{22}$$

where $T \oplus \delta\boldsymbol{\xi} = T \cdot \mathrm{Exp}(\delta\boldsymbol{\xi})$ is the $SE(3)$ right-plus and $+$ is standard vector addition ((Solà et al., 2021), Appendix E Eq. 186). The composite perturbation vector is:

$$\delta\mathbf{x} = \begin{pmatrix} \delta\boldsymbol{\xi} \\ \delta\mathbf{v} \\ \delta\boldsymbol{\omega} \\ \delta\mathbf{g}^B \\ \delta\mathbf{d}^B \\ \delta\mathbf{p}_1^B \\ \vdots \end{pmatrix} \in \mathbb{R}^{18+3N_f} \tag{23}$$

with $\delta\boldsymbol{\xi} = (\delta\boldsymbol{\rho},\, \delta\boldsymbol{\phi}) \in \mathbb{R}^6$ the $SE(3)$ tangent vector. The composite minus is the inverse: $\delta\mathbf{x} = \mathcal{X} \,\hat{\ominus}\, \hat{\mathcal{X}}$ gives $\delta\boldsymbol{\xi} = T \ominus \hat{T} = \mathrm{Log}(\hat{T}^{-1}T)$ for the $SE(3)$ block and $\delta\mathbf{v} = \mathbf{v} - \hat{\mathbf{v}}$ for the $\mathbb{R}^3$ blocks.

Composite derivative. Jacobians of functions on the composite state are computed block-wise ((Solà et al., 2021), Eq. 88–89): each block $\partial f_i / \partial \mathcal{X}_j$ uses the manifold derivative (Eq. 41a) for Lie group components and the standard derivative for $\mathbb{R}^3$ components. We denote this Jacobian as: $\frac{Df(\mathcal{X})}{D\mathcal{X}}$. This derivative coincides with the perturbation derivative, which in turn reduces to the standard partial derivative for Euclidean components.

2.2 Continuous-Discrete Extended Kalman Filter on a Composite Manifold State

The Extended Kalman Filter on a Composite Manifold State[1] maintains a Gaussian approximation of the state belief: at any time, the state estimate $\hat{\mathcal{X}}$ and covariance $\mathbf{P}$ represent a multivariate normal distribution $\mathcal{N}(\hat{\mathcal{X}}, \mathbf{P})$. This approximation is what justifies the use of chi-squared tests on innovations (Section 2.11).

The continuous-discrete formulation ((Crassidis and Junkins, 2004), Table 3.9) uses continuous-time dynamics with discrete-time measurements. We extend the Euclidean EKF formulation to a Composite Manifold state following ((Solà et al., 2021), Pages 1-12), (Huai and Gao, 2023), ((Bourmaud et al., 2014), Section 3) and ((Im, 2024), Section 5). We attempt to keep the notation similar to (Crassidis and Junkins, 2004) as this is what most are familiar with[2]:

$$\dot{\mathcal{X}}(t) = f(\mathcal{X}(t), \mathbf{u}(t), \mathbf{w}(t)) \tag{24}$$

$$\tilde{\mathbf{y}}(t_k) = h(\mathcal{X}(t_k), \mathbf{v}(t_k)) \tag{25}$$

[1] Also known as Multiplicative EKF (MEKF) , Error State KF (ESKF), M-CD-EKF, ...
[2] Our formulation is more of an Ansatz then something rigorously proven.

where:

$$f,\ h \quad \text{Continuously differentiable}$$
$$f \quad \text{System model}, \quad h \quad \text{Measurement model}$$
$$\mathcal{X}(t) \quad \text{True State}, \quad \mathbf{u}(t) \quad \text{Control input}$$
$$\mathbf{w}(t) \sim \mathcal{N}(\mathbf{0}, \mathcal{Q}(t)), \quad \mathbf{v}(t_k) \sim \mathcal{N}(\mathbf{0}, \mathbf{R}(t_k)) \quad \text{Uncorrelated noise}$$
$$\delta\mathbf{x} = \mathcal{X} \mathbin{\hat{\ominus}} \hat{\mathcal{X}} \quad \text{Estimation error}$$

2.2.1 Filter Equations

Initialise.

$$\hat{\mathcal{X}}(t_0) = \hat{\mathcal{X}}_0, \qquad \mathbf{P}_0 = \mathrm{E}\Big[(\mathcal{X}_0 \mathbin{\hat{\ominus}} \hat{\mathcal{X}}_0)\,(\mathcal{X}_0 \mathbin{\hat{\ominus}} \hat{\mathcal{X}}_0)^\top\Big] \tag{26}$$

Gain.

$$\mathbf{K}_k = \mathbf{P}_k^-\,\mathbf{H}_k^\top \left[\mathbf{H}_k\,\mathbf{P}_k^-\,\mathbf{H}_k^\top + \mathbf{H}_\mathbf{v}\mathbf{R}(t_k)\mathbf{H}_\mathbf{v}^\top\right]^{-1} \tag{27}$$

$$\mathbf{H}_k := \left.\frac{Dh}{D\mathcal{X}}\right|_{\hat{\mathcal{X}}_k^-,\ \mathbf{v}(t_k)=\mathbf{0}}, \quad \mathbf{H}_\mathbf{v} = \left.\frac{\partial h}{\partial \mathbf{v}}\right|_{\hat{\mathcal{X}}_k^-,\ \mathbf{v}(t_k)=\mathbf{0}} \tag{28}$$

Update.

$$\hat{\mathcal{X}}_k^+ = \hat{\mathcal{X}}_k^- \mathbin{\hat{\oplus}} \mathbf{K}_k \left[\tilde{\mathbf{y}}_k \mathbin{\hat{\ominus}} h(\hat{\mathcal{X}}_k^-, \mathbf{0})\right] \tag{29}$$

$$\mathbf{P}_k^+ = [\mathbf{I} - \mathbf{K}_k\,\mathbf{H}_k]\,\mathbf{P}_k^- \tag{30}$$

Propagation (continuous form).

$$\dot{\hat{\mathcal{X}}}(t) = f(\hat{\mathcal{X}}(t), \mathbf{u}(t), \mathbf{0}) \tag{31}$$

$$\dot{\mathbf{P}}(t) = F(t)\,\mathbf{P}(t) + \mathbf{P}(t)\,F^\top(t) + G(t)\,\mathcal{Q}(t)\,G^\top(t) \tag{32}$$

$$F(t) := \left.\frac{Df}{D\mathcal{X}}\right|_{\hat{\mathcal{X}}(t),\ \mathbf{u}(t),\ \mathbf{w}=\mathbf{0}}, \quad G(t) = \left.\frac{\partial f}{\partial \mathbf{w}}\right|_{\hat{\mathcal{X}}(t),\ \mathbf{u}(t),\ \mathbf{w}=\mathbf{0}} \tag{33}$$

Propagation (Discrete form). First-order Forward Euler integration of the estimate, component-wise according to the system model, with discrete-time process noise covariance approximated following ((Crassidis and Junkins, 2004), Equations 3.179, 3.193).

$$\hat{\mathcal{X}}_{k+1}^- = \hat{\mathcal{X}}_k^+ \mathbin{\hat{\oplus}} \Delta t_k \cdot f(\hat{\mathcal{X}}_k^+,\ \mathbf{u}_k, \mathbf{0}) \tag{34}$$

$$\Phi_k = \mathbf{I} + \Delta t_k \cdot F_k \tag{35}$$

$$\mathbf{P}_{k+1}^- = \Phi_k\,\mathbf{P}_k^+\,\Phi_k^\top + \Delta t_k\,G_k\,\mathcal{Q}_k\,G_k^\top \tag{36}$$

where $\Delta t_k = t_{k+1} - t_k$ is the actual frame interval (not assumed constant).

Note, state propagation being a left or right perturbation depends on the dynamics. Our equation is therefore strictly wrong, as whether it is a right or left perturbation is dictated by the dynamics and is a per sub-state decision. In our case we'll see the pose dynamics are left acting and use the left perturbation. You can map a left perturbation to a right perturbation using the adjoint. Thereby restating all perturbations as right acting.

2.2.2 State Augmentation and Marginalisation

The state vector has variable length: feature points enter and leave the filter as they are detected, lost, or reclassified. Three operations are needed (Mourikis and Roumeliotis, 2007).

Adding a state (augmentation). When a new feature with estimated position $\hat{\mathbf{p}}^B_{\text{new}}$ is added, the state and covariance grow:

$$\hat{\mathcal{X}} \leftarrow \langle \hat{\mathcal{X}},\ \hat{\mathbf{p}}^B_{\text{new}} \rangle, \qquad \mathbf{P} \leftarrow \begin{pmatrix} \mathbf{P} & \mathbf{P}\,\mathbf{J}_{\text{aug}}^\top \\ \mathbf{J}_{\text{aug}}\,\mathbf{P} & \mathbf{\Sigma}_{\text{new}} \end{pmatrix} \tag{37}$$

where $\mathbf{J}_{\text{aug}}$ captures how the new point's position depends on the current error state, and $\mathbf{\Sigma}_{\text{new}}$ is the full covariance of the point including measurement noise (Section 2.9). The blocks $\mathbf{P}\,\mathbf{J}_{\text{aug}}^\top$, $\mathbf{J}_{\text{aug}}\,\mathbf{P}$ encode the initial cross-correlations between the new point and all existing states. The same distinction as for F and H applies: the $\mathbb{R}^3$ columns of $\mathbf{J}_{\text{aug}}$ are standard partial derivatives; the $SE(3)$ column uses the right Jacobian ((Solà et al., 2021), Eq. 41a).

The position of augmented feature points depends on the estimated pose change, which itself depends on the filter state. Tracing $\mathbf{J}_{\text{aug}}$ through the full chain (state $\rightarrow$ pose $\rightarrow$ pose change $\rightarrow$ solver $\rightarrow$ reconstruction) is complex. We set $\mathbf{J}_{\text{aug}} = \delta\,\mathbf{1}$ with a small $\delta > 0$ to seed nonzero cross-correlations, avoiding the transient of recovering from exactly zero correlation[3].

Removing a state (marginalisation). When a feature is removed (lost, reclassified, or retired), delete the corresponding 3 rows and 3 columns from both $\hat{\mathcal{X}}$ and $\mathbf{P}$. No information about the remaining states is lost; their estimates and mutual cross-correlations are preserved.

Visibility transition (stereo $\leftrightarrow$ mono). A stereo feature demoted to mono, or a mono feature promoted to stereo, does not change its state representation. $\mathbf{p}^B_i \in \mathbb{R}^3$ is the same regardless of visibility class. The transition affects only:

- Which cameras produce measurements for this point (the measurement model row count changes).
- The group label (S, L, R) used for bookkeeping.

No augmentation or marginalisation is needed for visibility transitions.

[3] $\mathbf{I}$ is the identity. $\mathbf{1}$ is the matrix of only ones.

2.3 Projective Geometry

See (Szeliski, 2022) and (Ma et al., 2004) for comprehensive treatments.

2.3.1 Pinhole Camera Model

A 3D point passes through several transformations to become a pixel.

Homogeneous coordinates.

$$\tilde{\cdot} : \mathbf{p} \mapsto \begin{pmatrix} \mathbf{p} \\ 1 \end{pmatrix}, \quad (\tilde{\cdot})^{-1} : \begin{pmatrix} \mathbf{p} \\ s \end{pmatrix} \mapsto \frac{\mathbf{p}}{s} \qquad \text{Homogeneous representation} \tag{38}$$

Perspective divide. Projects a 3D camera-frame point to normalised image coordinates, discarding depth:

$$\pi_n(\mathbf{p}) = \frac{1}{p_z} \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} x_n \\ y_n \end{pmatrix} \tag{39}$$

Lens distortion. Maps undistorted normalised coordinates to distorted normalised coordinates. Model-dependent (Brown-Conrady, equidistant, etc.):

$$D : \mathbb{R}^2 \to \mathbb{R}^2, \quad D(x_n, y_n) = (x_d, y_d) \tag{40}$$

Camera intrinsics. Maps distorted normalised coordinates to pixel coordinates:

$$\mathbf{K} = \begin{pmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{pmatrix}, \qquad \kappa(x_d, y_d) = \begin{pmatrix} f_u\, x_d + c_u \\ f_v\, y_d + c_v \end{pmatrix} \tag{41}$$

Full projection. From camera frame to pixel:

$$\boldsymbol{\pi} = \kappa \circ D \circ \pi_n : \mathbb{R}^3 \to \mathbb{R}^2 \tag{42}$$

Inverse projection. From pixel to normalised coordinates and then to a ray (up to unknown depth Z):

$$(u, v) \xrightarrow{\kappa^{-1}} (x_d, y_d) \xrightarrow{D^{-1}} (x_n, y_n) \xrightarrow{\times Z} \mathbf{p}^c = Z \begin{pmatrix} x_n \\ y_n \\ 1 \end{pmatrix} \tag{43}$$

The depth Z cannot be recovered from a single image; projection is a many-to-one mapping.

2.3.2 Camera Extrinsics

Rotation convention. $\mathbf{R}_{AB}$ rotates vectors from frame B into frame A: $\mathbf{v}^A = \mathbf{R}_{AB}\,\mathbf{v}^B$. Frames: B_0 (initial body / world), B (body), L (left camera), R (right camera).

Body to camera. A body-frame point $\mathbf{p}^B$ in camera $c \in \{L, R\}$:

$$\mathbf{p}^c = \mathbf{R}_{cB}\,\mathbf{p}^B + \mathbf{t}_c \tag{44}$$

The full projection from body frame to pixel:

$$\mathbf{u} = \boldsymbol{\pi}\Big(\mathbf{R}_{cB}\,\mathbf{p}^B + \mathbf{t}_c\Big) \tag{45}$$

Camera to camera. For canonical stereo with baseline $b = \|\mathbf{t}_R - \mathbf{t}_L\|$:

$$\mathbf{p}^R = \mathbf{R}_{RL}\,\mathbf{p}^L + \mathbf{t}_{RL} \tag{46}$$

For parallel cameras: $\mathbf{R}_{RL} = \mathbf{I}$, $\mathbf{t}_{RL} = (-b, 0, 0)^\top$.

2.3.3 Stereo Geometry

For a rectified stereo pair, corresponding points lie on the same horizontal scanline (epipolar line). The horizontal disparity encodes depth:

$$d = u_L - u_R \qquad \text{Disparity (pixels)} \tag{47}$$

$$Z = \frac{f_u \cdot b}{d} \qquad \text{Depth from disparity} \tag{48}$$

Depth uncertainty from first-order propagation of disparity noise σ_d:

$$\sigma_Z = \left|\frac{\partial Z}{\partial d}\right| \sigma_d = \frac{f_u\, b}{d^2}\,\sigma_d = \frac{Z^2}{f_u\, b}\,\sigma_d \tag{49}$$

Depth uncertainty grows quadratically with distance.

2.3.4 Epipolar Constraint

Given a point in the left image at pixel (u_L, v_L), its correspondence in the right image is constrained to lie on a line (the epipolar line). For a rectified pair this is simply $v_R = v_L$, the same row. This reduces stereo matching from a 2D search to a 1D search along the scanline.

2.4 Rigid Body Dynamics

These equations can among others be found in (Beard and McLain, 2012).

$$\sum \mathbf{F} = m\dot{\mathbf{v}} \qquad \text{Newton's second law} \tag{50}$$

$$(\dot{\mathbf{v}})_{\text{inertial}} = (\dot{\mathbf{v}})_{\text{body}} + \boldsymbol{\omega} \times \mathbf{v} \qquad \text{Transport theorem} \tag{51}$$

$$\mathbf{J}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) = \boldsymbol{\tau} \qquad \text{Euler's rotation equations} \tag{52}$$

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \qquad \text{Torque} \tag{53}$$

$$\mathbf{F}_{\text{drag}} = -C_d\,\mathbf{v} \qquad \text{Linear drag} \tag{54}$$

$$T = k_f\,\omega_r^2 \qquad \text{Propeller thrust} \tag{55}$$

$$Q = k_m\,\omega_r^2 \qquad \text{Reactive propeller torque} \tag{56}$$

$$[\mathbf{a}]_\times = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix} \qquad \text{Skew-symmetric (cross-product) matrix} \tag{57}$$

Transport theorem for world-fixed vectors. A vector $\mathbf{a}^{B_0}$ constant in the B_0 frame, expressed in the rotating body frame as $\mathbf{a}^B = \mathbf{R}_{B,B_0}\,\mathbf{a}^{B_0}$, satisfies:

$$\dot{\mathbf{a}}^B = -\boldsymbol{\omega} \times \mathbf{a}^B \tag{58}$$

Derived from $\dot{\mathbf{R}}_{B,B_0} = -[\boldsymbol{\omega}]_\times\,\mathbf{R}_{B,B_0}$ (Academic Flight, n.d.).

2.5 System Model

We derive the system model $f(\mathcal{X}, \mathbf{u}, \mathbf{w})$ for the continuous-discrete filter (Section 2.2), followed by the process noise covariance $\mathcal{Q}$, the error dynamics Jacobian F, and the noise Jacobian G.

2.5.1 State Vector

The composite manifold state (Section 2.1.1) is:

$$\mathcal{X} = \left\langle\, T,\ \mathbf{v},\ \boldsymbol{\omega},\ \mathbf{g}^B,\ \mathbf{d}^B,\ \mathbf{p}_1^B,\ \ldots,\ \mathbf{p}_{N_f}^B \,\right\rangle \tag{59}$$

where $T = T_{B_t,B_0} \in SE(3)$ is the body pose (Section 2.1) and all other components are in $\mathbb{R}^3$. The pose acts as $\mathbf{p}^{B_t} = \mathbf{R}_{B_t}\,\mathbf{p}^{B_0} + \mathbf{t}_{B_t}$, where $\mathbf{t}_{B_t}$ is the B_0 origin expressed in the body frame, a world-fixed point, identical in nature to the feature positions $\mathbf{p}_i^B$.

The composite perturbation vector (Section 2.1.1, Equation 23) is:

$$\delta\mathbf{x} = \begin{pmatrix} \delta\boldsymbol{\xi} \\ \delta\mathbf{v} \\ \delta\boldsymbol{\omega} \\ \delta\mathbf{g}^B \\ \delta\mathbf{d}^B \\ \delta\mathbf{p}_1^B \\ \vdots \\ \delta\mathbf{p}_{N_f}^B \end{pmatrix} \in \mathbb{R}^{18+3N_f} \tag{60}$$

with $\delta\boldsymbol{\xi} = (\delta\boldsymbol{\rho},\, \delta\boldsymbol{\phi}) \in \mathbb{R}^6$ the $SE(3)$ tangent vector and $N_f = N_S + N_L + N_R$. The $\mathbb{R}^3$ components are:

$$\mathbf{v} \in \mathbb{R}^3 \qquad \text{Linear velocity in body frame} \tag{61}$$

$$\boldsymbol{\omega} \in \mathbb{R}^3 \qquad \text{Angular velocity in body frame} \tag{62}$$

$$\mathbf{g}^B \in \mathbb{R}^3 \qquad \text{Gravity in body frame } (\|\mathbf{g}^B\| = g) \tag{63}$$

$$\mathbf{d}^B \in \mathbb{R}^3 \qquad \text{Wind/disturbances in body frame} \tag{64}$$

Feature positions partitioned by visibility:

$$\mathbf{p}_j^S \in \mathbb{R}^3,\ j = 1, \ldots, N_S \qquad \text{Stereo features} \tag{65}$$

$$\mathbf{p}_j^L \in \mathbb{R}^3,\ j = 1, \ldots, N_L \qquad \text{Mono-left features} \tag{66}$$

$$\mathbf{p}_j^R \in \mathbb{R}^3,\ j = 1, \ldots, N_R \qquad \text{Mono-right features} \tag{67}$$

All positions are in body frame. The visibility label (S, L, R) determines which measurements a point contributes (Section 2.6).

Gravity is split from disturbances: gravity is well approximated as constant in the world frame with known magnitude, requiring only small process noise. Wind is genuinely stochastic, requiring larger process noise. Separate tuning avoids compromising one to accommodate the other.

2.5.2 Input Vector

$$\mathbf{u} = \begin{pmatrix} T_1 & T_2 & T_3 & T_4 \end{pmatrix}^\top \in \mathbb{R}^4 \tag{68}$$

where $T_i = k_f\,\omega_{r,i}^2$ is the thrust from rotor i.

2.5.3 Dynamics

Block 0: Pose ($\dot{T}$). The body-frame velocity twist $\boldsymbol{\xi}_v = (\mathbf{v},\, \boldsymbol{\omega})$ drives the pose. The continuous dynamics are:

$$\boxed{\dot{T} = -\boldsymbol{\xi}_v^\wedge\, T} \tag{69}$$

Derivation. Write $T = [\mathbf{R},\, \mathbf{t};\, \mathbf{0}^\top,\, 1]$ and compute the right-hand side using the Lie algebra element $\boldsymbol{\xi}_v^\wedge = [[\boldsymbol{\omega}]_\times,\, \mathbf{v};\, \mathbf{0}^\top,\, 0]$ (Section 2.1):

$$-\boldsymbol{\xi}_v^\wedge T = -\begin{pmatrix} [\boldsymbol{\omega}]_\times & \mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix} \begin{pmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{pmatrix} = \begin{pmatrix} -[\boldsymbol{\omega}]_\times \mathbf{R} & -[\boldsymbol{\omega}]_\times \mathbf{t} - \mathbf{v} \\ \mathbf{0}^\top & 0 \end{pmatrix} \tag{70}$$

Reading off the blocks: $\dot{\mathbf{R}} = -[\boldsymbol{\omega}]_\times \mathbf{R}$, which is the rotation kinematics (Section 2.4), and $\dot{\mathbf{t}} = -\mathbf{v} - [\boldsymbol{\omega}]_\times \mathbf{t} = -\mathbf{v} - \boldsymbol{\omega} \times \mathbf{t}$, which is the transport theorem (Equation 58) applied to $\mathbf{t}_{B_t}$, the B_0 origin expressed in body frame, a world-fixed point. Both known component dynamics are recovered.

No process noise enters the pose directly: pose uncertainty is inherited entirely from velocity and angular velocity through integration.

Propagation (left multiply, see Section 2.2):

$$\hat{T}_{k+1}^- = \operatorname{Exp}(-\hat{\boldsymbol{\xi}}_{v,k}\, \Delta t_k) \cdot \hat{T}_k^+ \tag{71}$$

This follows from the ODE $\dot{T} = AT$ with constant $A = -\boldsymbol{\xi}_v^\wedge$, whose solution is $T(t + \Delta t) = \exp(A\Delta t)\, T(t) = \operatorname{Exp}(-\boldsymbol{\xi}_v \Delta t) \cdot T(t)$.

Block 1: Translational ($\dot{\mathbf{v}}$). Newton's second law in the rotating body frame:

$$\boxed{\dot{\mathbf{v}} = \frac{1}{m}\mathbf{F}_{\text{thrust}}(\mathbf{u} + \mathbf{w}_u) - \boldsymbol{\omega} \times \mathbf{v} - \frac{C_d}{m}\mathbf{v} + \mathbf{g}^B + \mathbf{d}^B} \tag{72}$$

with $\mathbf{F}_{\text{thrust}} = (0, 0, -\sum(T_i + w_{u,i}))^\top$ in FRD. Actuator noise $\mathbf{w}_u$ enters through the thrust mapping $B_{\mathbf{v}}$ (Equation 83).

Block 2: Rotational ($\dot{\boldsymbol{\omega}}$). Euler's rotation equations, assuming the inertia tensor is diagonal $\mathbf{J} = \operatorname{diag}(J_{xx}, J_{yy}, J_{zz})$ (valid for a symmetric X-configuration where the body axes align with the principal axes of inertia):

$$\boxed{\dot{\boldsymbol{\omega}} = \mathbf{J}^{-1}\left(\boldsymbol{\tau}(\mathbf{u} + \mathbf{w}_u) - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega})\right) + \mathbf{w}_\omega} \tag{73}$$

Actuator noise $\mathbf{w}_u$ enters through the torque mapping B_ω (Equation 84). The additional process noise $\mathbf{w}_\omega$ accounts for unmodelled aerodynamic torques (e.g. asymmetric wind loading) that are not captured by the actuator channel.

Torques from the X-configuration mixing matrix (Noise adds to each T_i, omitted here for clarity):

$$\tau_1 = \tfrac{l}{\sqrt{2}}(T_4 + T_3 - T_1 - T_2) \qquad \text{Roll} \tag{74}$$

$$\tau_2 = \tfrac{l}{\sqrt{2}}(T_1 + T_4 - T_2 - T_3) \qquad \text{Pitch} \tag{75}$$

$$\tau_3 = \tfrac{k_m}{k_f}(T_2 + T_4 - T_1 - T_3) \qquad \text{Yaw} \tag{76}$$

Block 3: Gravity ($\dot{\mathbf{g}}^B$). From the transport theorem (Equation 58):

$$\boxed{\dot{\mathbf{g}}^B = -\boldsymbol{\omega} \times \mathbf{g}^B + \mathbf{w}_g} \tag{77}$$

Process noise $\mathbf{w}_g$ accounts for the approximation that gravity is constant in the world frame (small spectral density $\mathcal{Q}_g$).

Block 4: Disturbances ($\dot{\mathbf{d}}^B$). Same transport equation:

$$\boxed{\dot{\mathbf{d}}^B = -\boldsymbol{\omega} \times \mathbf{d}^B + \mathbf{w}_d} \tag{78}$$

Process noise $\mathbf{w}_d$ drives the disturbance as a random walk in the world frame (larger spectral density $\mathcal{Q}_d$).

Block 5: Feature positions ($\dot{\mathbf{p}}_i^B$). A world-stationary point in the body frame satisfies (derived from $\mathbf{p}_i^B = \mathbf{R}_{B,B_0}(\mathbf{p}_i^{B_0} - \mathbf{o}^{B_0})$ with $\dot{\mathbf{p}}_i^{B_0} = \mathbf{0}$):

$$\boxed{\dot{\mathbf{p}}_i^B = -\mathbf{v} - \boldsymbol{\omega} \times \mathbf{p}_i^B} \tag{79}$$

Identical for all visibility groups. Points are assumed stationary in the world frame; violations of this assumption are detected by the chi-squared test (Section 2.11) and the point is removed. Note that the translation component $\mathbf{t}_{B_t}$ of the pose satisfies the same equation (Block 0 derivation above).

2.5.4 Process Noise

The process noise vector is:

$$\mathbf{w} = \begin{pmatrix} \mathbf{w}_u \\ \mathbf{w}_\omega \\ \mathbf{w}_g \\ \mathbf{w}_d \end{pmatrix} \in \mathbb{R}^{13} \tag{80}$$

where $\mathbf{w}_u \in \mathbb{R}^4$ is actuator noise on the rotor thrusts, $\mathbf{w}_\omega \in \mathbb{R}^3$ is unmodelled aerodynamic torque, and the remaining terms are as before. The continuous-time spectral density (Section 2.2):

$$\mathcal{Q} = \mathrm{diag}(\mathcal{Q}_u,\ \mathcal{Q}_\omega,\ \mathcal{Q}_g,\ \mathcal{Q}_d) \in \mathbb{R}^{13\times 13} \tag{81}$$

where $\mathcal{Q}_u \in \mathbb{R}^{4\times 4}$ is the actuator spectral density and $\mathcal{Q}_\omega$, $\mathcal{Q}_g$, $\mathcal{Q}_d$ are each 3×3 (units: state^2/time). Actuator noise enters both the velocity and angular velocity through the thrust and torque mappings. No noise enters the pose (T) or the linear velocity ($\mathbf{v}$) directly beyond the actuator channel: pose uncertainty is inherited from $\mathbf{v}$ and $\boldsymbol{\omega}$ through integration, and velocity uncertainty is inherited from the forcing terms $\mathbf{g}^B$, $\mathbf{d}^B$, $\boldsymbol{\omega}$, and the actuators through the dynamics (Equation 72).

2.5.5 Complete System Model

The $SE(3)$ dynamics are $\dot{T} = -\boldsymbol{\xi}_v^\wedge T$ (Equation 69). The $\mathbb{R}^3$ dynamics are assembled as:

$$\dot{\mathbf{x}}_\mathbb{R} = f_\mathbb{R}(\mathcal{X}, \mathbf{u}, \mathbf{w}) = \begin{pmatrix} \frac{1}{m}\mathbf{F}_{\text{thrust}}(\mathbf{u} + \mathbf{w}_u) - \boldsymbol{\omega} \times \mathbf{v} - \frac{C_d}{m}\mathbf{v} + \mathbf{g}^B + \mathbf{d}^B \\ \mathbf{J}^{-1}(\boldsymbol{\tau}(\mathbf{u} + \mathbf{w}_u) - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega})) + \mathbf{w}_\omega \\ -\boldsymbol{\omega} \times \mathbf{g}^B + \mathbf{w}_g \\ -\boldsymbol{\omega} \times \mathbf{d}^B + \mathbf{w}_d \\ -\mathbf{v} - \boldsymbol{\omega} \times \mathbf{p}_1^B \\ \vdots \\ -\mathbf{v} - \boldsymbol{\omega} \times \mathbf{p}_{N_f}^B \end{pmatrix} \tag{82}$$

Together with $\dot{T}$ these form the complete system model $f(\mathcal{X}, \mathbf{u}, \mathbf{w})$. Setting $\mathbf{w} = \mathbf{0}$ recovers the nominal dynamics used for state propagation (Section 2.2).

2.5.6 Noise Jacobian G

The noise Jacobian $G = \partial f / \partial \mathbf{w}$ is the standard partial derivative (Section 2.1.1) since $\mathbf{w} \in \mathbb{R}^{13}$ is Euclidean. Define the actuator-to-force and actuator-to-torque mappings:

$$B_\mathbf{v} = \frac{1}{m}\frac{\partial \mathbf{F}_{\text{thrust}}}{\partial \mathbf{u}} = \frac{1}{m}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 \end{pmatrix} \in \mathbb{R}^{3\times 4} \tag{83}$$

$$B_\omega = \mathbf{J}^{-1}\frac{\partial \boldsymbol{\tau}}{\partial \mathbf{u}} = \mathbf{J}^{-1}\begin{pmatrix} -\frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} \\ \frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} \\ -\frac{k_m}{k_f} & \frac{k_m}{k_f} & -\frac{k_m}{k_f} & \frac{k_m}{k_f} \end{pmatrix} \in \mathbb{R}^{3\times 4} \tag{84}$$

The full G matrix maps the (13)-dimensional noise into the $(18 + 3N_f)$-dimensional perturbation space. Column groups correspond to $\mathbf{w}_u$, $\mathbf{w}_\omega$, $\mathbf{w}_g$, $\mathbf{w}_d$:

$$G = \begin{pmatrix} \mathbf{0}_{6\times 4} & \mathbf{0}_{6\times 3} & \mathbf{0}_{6\times 3} & \mathbf{0}_{6\times 3} \\ B_\mathbf{v} & \mathbf{0}_{3\times 3} & \mathbf{0} & \mathbf{0} \\ B_\omega & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} \\ \mathbf{0}_{3\times 4} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0}_{3\times 4} & \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \\ \mathbf{0}_{3\times 4} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \vdots & & & \vdots \\ \mathbf{0}_{3\times 4} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \in \mathbb{R}^{(18+3N_f)\times 13} \tag{85}$$

Row groups correspond to $\delta\boldsymbol{\xi}$, $\delta\mathbf{v}$, $\delta\boldsymbol{\omega}$, $\delta\mathbf{g}^B$, $\delta\mathbf{d}^B$, $\delta\mathbf{p}_i^B$. The pose row is zero (no direct noise). The velocity row receives actuator noise through $B_\mathbf{v}$. The angular velocity row receives both actuator noise through B_ω (structured, via the mixing matrix) and unmodelled torque $\mathbf{w}_\omega$ (unstructured).

2.5.7 Error Dynamics Jacobian F

F is the composite Jacobian (Section 2.1.1) of the error dynamics $\dot{\delta\mathbf{x}} \approx F\,\delta\mathbf{x}$, evaluated at $\mathbf{w} = \mathbf{0}$. The noise terms do not affect F: they enter linearly through G and do not couple to the state.

$\mathbb{R}^3$ blocks. For the $\mathbb{R}^3$ states, F is obtained by standard differentiation of $f_\mathbb{R}$ with $\mathbf{w} = \mathbf{0}$ ((Crassidis and Junkins, 2004), Eqs. 3.239–3.244):

$$F_{vv} = -[\hat{\boldsymbol{\omega}}]_\times - \tfrac{C_d}{m} I_3 \qquad F_{v\omega} = [\hat{\mathbf{v}}]_\times \tag{86}$$

$$F_{\omega\omega} = \mathbf{J}^{-1}([\mathbf{J}\hat{\boldsymbol{\omega}}]_\times - [\hat{\boldsymbol{\omega}}]_\times \mathbf{J}) \tag{87}$$

$$F_{g\omega} = [\hat{\mathbf{g}}^B]_\times \qquad F_{gg} = -[\hat{\boldsymbol{\omega}}]_\times \tag{88}$$

$$F_{d\omega} = [\hat{\mathbf{d}}^B]_\times \qquad F_{dd} = -[\hat{\boldsymbol{\omega}}]_\times \tag{89}$$

$$F_{pv} = -I_3 \qquad F_{p_i\omega} = [\hat{\mathbf{p}}_i^B]_\times \tag{90}$$

$$F_{p_ip_i} = -[\hat{\boldsymbol{\omega}}]_\times \tag{91}$$

Cross-point terms vanish: $\partial\dot{\mathbf{p}}_i^B/\partial\mathbf{p}_j^B = \mathbf{0}$ for $j \neq i$.

$SE(3)$ blocks. For the pose, the error dynamics are $\dot{\delta\boldsymbol{\xi}} = -\mathrm{Ad}_{\hat{T}^{-1}}(\delta\mathbf{v},\, \delta\boldsymbol{\omega})^\top$.

Derivation. The true pose satisfies $\dot{T}_\text{true} = -(\boldsymbol{\xi}_v + \delta\boldsymbol{\xi}_v)^\wedge T_\text{true}$ where $\delta\boldsymbol{\xi}_v = (\delta\mathbf{v},\, \delta\boldsymbol{\omega})$. With $T_\text{true} = \hat{T} \cdot \mathrm{Exp}(\delta\boldsymbol{\xi})$ ((Solà et al., 2021), Eq. 25), expand the left side via the product rule:

$$\dot{\hat{T}}\,\mathrm{Exp}(\delta\boldsymbol{\xi}) + \hat{T}\,\tfrac{d}{dt}\mathrm{Exp}(\delta\boldsymbol{\xi}) = -(\boldsymbol{\xi}_v + \delta\boldsymbol{\xi}_v)^\wedge\,\hat{T}\,\mathrm{Exp}(\delta\boldsymbol{\xi}) \tag{92}$$

The estimate dynamics $\dot{\hat{T}} = -\boldsymbol{\xi}_v^\wedge \hat{T}$ (Equation 69) cancel from both sides:

$$\hat{T}\,\tfrac{d}{dt}\mathrm{Exp}(\delta\boldsymbol{\xi}) = -(\delta\boldsymbol{\xi}_v)^\wedge\,\hat{T}\,\mathrm{Exp}(\delta\boldsymbol{\xi}) \tag{93}$$

Left-multiply by $\hat{T}^{-1}$ and apply the adjoint conjugation property $\mathcal{X}^{-1}\boldsymbol{\tau}^\wedge\mathcal{X} = (\mathrm{Ad}_{\mathcal{X}^{-1}}\boldsymbol{\tau})^\wedge$ ((Solà et al., 2021), Eq. 29):

$$\tfrac{d}{dt}\mathrm{Exp}(\delta\boldsymbol{\xi}) = -(\mathrm{Ad}_{\hat{T}^{-1}}\,\delta\boldsymbol{\xi}_v)^\wedge\,\mathrm{Exp}(\delta\boldsymbol{\xi}) \tag{94}$$

First order: $\mathrm{Exp}(\delta\boldsymbol{\xi}) \approx \mathcal{E} + \delta\boldsymbol{\xi}^\wedge$ ((Solà et al., 2021), Eq. 16, truncated after the first term), so $\tfrac{d}{dt}\mathrm{Exp}(\delta\boldsymbol{\xi}) \approx (\dot{\delta\boldsymbol{\xi}})^\wedge$ and $\mathrm{Exp}(\delta\boldsymbol{\xi}) \approx \mathcal{E}$ on the right. Extracting the vector:

$$\dot{\delta\boldsymbol{\xi}} = -\mathrm{Ad}_{\hat{T}^{-1}}\,\delta\boldsymbol{\xi}_v = -\mathrm{Ad}_{\hat{T}^{-1}}\begin{pmatrix}\delta\mathbf{v}\\ \delta\boldsymbol{\omega}\end{pmatrix} \tag{95}$$

The resulting F blocks are:

$$F_{\delta\boldsymbol{\xi},\,\delta\boldsymbol{\xi}} = \mathbf{0}_{6\times 6}, \qquad F_{\delta\boldsymbol{\xi},\,(\delta\mathbf{v},\,\delta\boldsymbol{\omega})} = -\mathrm{Ad}_{\hat{T}^{-1}} \tag{96}$$

with $\mathrm{Ad}_{\hat{T}^{-1}}$ from Equation 20 evaluated at $\hat{T}^{-1}$. All other entries in the $\delta\boldsymbol{\xi}$ row are zero ($\delta\mathbf{g}$, $\delta\mathbf{d}$, $\delta\mathbf{p}_i$ do not enter the pose error dynamics). All entries in the $\delta\boldsymbol{\xi}$ column from existing rows are zero (no $\mathbb{R}^3$ dynamics depend on pose).

Full F matrix. Column order: $\delta\boldsymbol{\xi}$, $\delta\mathbf{v}$, $\delta\boldsymbol{\omega}$, $\delta\mathbf{g}^B$, $\delta\mathbf{d}^B$, $\delta\mathbf{p}_1^B$, $\ldots$, $\delta\mathbf{p}_{N_f}^B$:

$$F = \left(\begin{array}{c|cc|cc|ccc}
\mathbf{0}_6 & \multicolumn{2}{c|}{-\mathrm{Ad}_{\hat{T}^{-1}}} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\
\hline
\mathbf{0} & F_{vv} & F_{v\omega} & I_3 & I_3 & \mathbf{0} & \cdots & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & F_{\omega\omega} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\
\hline
\mathbf{0} & \mathbf{0} & F_{g\omega} & F_{gg} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & F_{d\omega} & \mathbf{0} & F_{dd} & \mathbf{0} & \cdots & \mathbf{0} \\
\hline
\mathbf{0} & F_{pv} & F_{p_1\omega} & \mathbf{0} & \mathbf{0} & F_{p_1p_1} & \cdots & \mathbf{0} \\
\vdots & \vdots & \vdots & & & & \ddots & \\
\mathbf{0} & F_{pv} & F_{p_{N_f}\omega} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & F_{p_{N_f}p_{N_f}}
\end{array}\right) \tag{97}$$

The lower-right block is block-diagonal in the per-point states: each point i has its own diagonal block $F_{p_ip_i} = -[\hat{\boldsymbol{\omega}}]_\times$ and off-diagonal coupling $F_{p_i\omega} = [\hat{\mathbf{p}}_i^B]_\times$ to angular velocity. All points share the same $F_{pv} = -I_3$. Cross-point terms vanish: $\partial\dot{\mathbf{p}}_i^B/\partial\mathbf{p}_j^B = \mathbf{0}$ for $j \neq i$. The first row (pose error dynamics) couples to velocity and angular velocity through $-\mathrm{Ad}_{\hat{T}^{-1}}$. The first column is zero throughout (no $\mathbb{R}^3$ dynamics depend on pose).

2.5.8 Initialisation

$$\hat{T}_0 = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{pmatrix} \qquad \text{Body at } B_0 \text{ origin, aligned} \tag{98}$$

$$\hat{\mathbf{v}}_0 = \mathbf{0} \qquad \text{Stationary start} \tag{99}$$

$$\hat{\boldsymbol{\omega}}_0 = \mathbf{0} \qquad \text{No initial rotation} \tag{100}$$

$$\hat{\mathbf{g}}_0^B = (0, 0, g)^\top \qquad \text{FRD: down} = +e_3 \tag{101}$$

$$\hat{\mathbf{d}}_0^B = -\hat{\mathbf{g}}_0^B \qquad \text{Normal force} \tag{102}$$

No feature points are present at initialisation. All points must pass the velocity entry test (Section 2.11) before being augmented into the filter. The first frame provides stereo and mono detections; these are tracked to the second frame where the joint solver (Section 2.8.7) produces position and velocity estimates with covariance, at which point they are augmented (Section 2.2.2). Initial covariance:

$$\mathbf{P}_0 = \mathrm{diag}\Big(\sigma_\xi^2 I_6,\ \sigma_v^2 I_3,\ \sigma_\omega^2 I_3,\ \sigma_g^2 I_3,\ \sigma_d^2 I_3\Big) \tag{103}$$

with no feature blocks until augmentation.

2.6 Measurement Model

We derive the measurement model $\tilde{\mathbf{y}}(t_k) = h(\mathcal{X}(t_k), \mathbf{v}(t_k))$ for the continuous-discrete filter (Section 2.2), followed by the measurement noise $\mathbf{R}$, and the Jacobians $\mathbf{H}_k$ and $\mathbf{H}_\mathbf{v}$.

2.6.1 Observations

At each frame t_k, the observations are LK-tracked pixel coordinates of feature points in one or both cameras:

$$\tilde{\mathbf{y}}(t_k) = \begin{pmatrix} \tilde{\mathbf{y}}_1 \\ \vdots \\ \tilde{\mathbf{y}}_{N_f} \end{pmatrix} \in \mathbb{R}^{\sum_i \nu_i} \tag{104}$$

where $\tilde{\mathbf{y}}_i \in \mathbb{R}^{\nu_i}$ are the pixel coordinates of point i, with $\nu_i = 4$ (stereo: left and right cameras) or $\nu_i = 2$ (mono: single camera).

2.6.2 Measurement Function

The measurement prediction maps the state to pixel coordinates via direct geometric projection (Equation 45). For a single point i in camera $c \in \{L, R\}$:

$$h_{i,c}(\mathcal{X}, \mathbf{v}_{i,c}) = \boldsymbol{\pi}(\mathbf{R}_{cB}\, \mathbf{p}_i^B + \mathbf{t}_c) + \mathbf{v}_{i,c} \tag{105}$$

where $\mathbf{v}_{i,c} \in \mathbb{R}^2$ is additive pixel noise in camera c for point i. The noise is additive because the dominant error source (LK sub-pixel localisation) acts directly on the pixel coordinates.

Stereo feature ($\mathcal{F}_S$). A stereo feature $\mathbf{p}_i^B$ is observed in both cameras:

$$h_{S,i}(\mathcal{X}, \mathbf{v}_i) = \begin{pmatrix} \boldsymbol{\pi}(\mathbf{R}_{LB}\, \mathbf{p}_i^B + \mathbf{t}_L) \\ \boldsymbol{\pi}(\mathbf{R}_{RB}\, \mathbf{p}_i^B + \mathbf{t}_R) \end{pmatrix} + \mathbf{v}_i = \begin{pmatrix} u_L \\ v_L \\ u_R \\ v_R \end{pmatrix} \in \mathbb{R}^4 \tag{106}$$

with $\mathbf{v}_i \in \mathbb{R}^4$.

Why not a separate disparity measurement? Disparity $d = u_L - u_R$ is a linear function of the four pixel coordinates already observed. Adding it as a 5th row adds zero information. A depth error produces different innovations in u_L and u_R because the Jacobians $\partial h / \partial \mathbf{p}_i^B$ differ between the two cameras (different extrinsics). The filter exploits this geometric discrepancy through the existing four rows.

Mono feature ($\mathcal{F}_M$). A mono feature is observed in one camera $c \in \{L, R\}$:

$$h_{M,i}(\mathcal{X}, \mathbf{v}_i) = \boldsymbol{\pi}(\mathbf{R}_{cB}\, \mathbf{p}_i^B + \mathbf{t}_c) + \mathbf{v}_i = \begin{pmatrix} u_c \\ v_c \end{pmatrix} \in \mathbb{R}^2 \tag{107}$$

with $\mathbf{v}_i \in \mathbb{R}^2$.

Assembled measurement function. Stacking all points:

$$h(\mathcal{X}, \mathbf{v}) = \begin{pmatrix} h_{\cdot,1}(\mathcal{X}, \mathbf{v}_1) \\ \vdots \\ h_{\cdot,N_f}(\mathcal{X}, \mathbf{v}_{N_f}) \end{pmatrix} \in \mathbb{R}^{\sum_i \nu_i} \tag{108}$$

where each $h_{\cdot,i}$ is $h_{S,i}$ or $h_{M,i}$ according to the visibility of point i.

2.6.3 Measurement Noise

The noise vector $\mathbf{v} = (\mathbf{v}_1, \ldots, \mathbf{v}_{N_f})^\top$ has covariance:

$$\mathbf{R} = \mathrm{diag}\Big(\sigma_{\mathrm{px}}^2\, \mathbf{I}_{\nu_1},\ \ldots,\ \sigma_{\mathrm{px}}^2\, \mathbf{I}_{\nu_{N_f}}\Big) = \sigma_{\mathrm{px}}^2\, \mathbf{I}_{\sum_i \nu_i} \tag{109}$$

where $\sigma_{\mathrm{px}} \approx 0.3$–$1.0$ px is a tuning parameter representing LK sub-pixel accuracy. All pixel measurements share the same noise level; the block-diagonal structure reflects independence between points and between cameras.

2.6.4 Measurement Jacobian $\mathbf{H}_k$

Following Section 2.2, $\mathbf{H}_k = \frac{Dh}{D\mathcal{X}}\Big|_{\hat{\mathcal{X}}_k^-,\, \mathbf{v}=\mathbf{0}}$ is the composite Jacobian (Section 2.1.1). Since the measurement noise is additive, setting $\mathbf{v} = \mathbf{0}$ simply removes it.

Projection Jacobian. For camera c, define $\mathbf{p}_i^c = \mathbf{R}_{cB}\, \mathbf{p}_i^B + \mathbf{t}_c$. The chain rule through $\boldsymbol{\pi} = \kappa \circ D \circ \pi_n$ (Section 2.3) gives:

$$\frac{\partial \boldsymbol{\pi}}{\partial \mathbf{p}^c} = \mathbf{K}_{2\times 2}\, \mathbf{J}_D\, \frac{1}{p_z^c} \begin{pmatrix} 1 & 0 & -x_n \\ 0 & 1 & -y_n \end{pmatrix} \tag{110}$$

$$\mathbf{K}_{2\times 2} = \begin{pmatrix} f_u & 0 \\ 0 & f_v \end{pmatrix}, \quad \mathbf{J}_D = \frac{\partial D}{\partial (x_n, y_n)} \tag{111}$$

The Jacobian with respect to the body-frame position:

$$\frac{\partial \boldsymbol{\pi}}{\partial \mathbf{p}_i^B}\bigg|_c = \frac{\partial \boldsymbol{\pi}}{\partial \mathbf{p}^c}\, \mathbf{R}_{cB} \tag{112}$$

Per-point row. The measurement depends only on $\mathbf{p}_i^B$ — not on T, $\mathbf{v}$, $\boldsymbol{\omega}$, $\mathbf{g}^B$, $\mathbf{d}^B$, or any other point $\mathbf{p}_j^B$ $(j \neq i)$. The full Jacobian row for point i in camera c:

$$\mathbf{H}_i^c = \begin{pmatrix} \mathbf{0}_{2\times 6} & \mathbf{0}_{2\times 3} & \mathbf{0}_{2\times 3} & \mathbf{0}_{2\times 3} & \mathbf{0}_{2\times 3} & \cdots & \frac{\partial \boldsymbol{\pi}}{\partial \mathbf{p}_i^B}\Big|_c & \cdots & \mathbf{0}_{2\times 3} \end{pmatrix} \tag{113}$$

Nonzero only in the columns corresponding to $\mathbf{p}_i^B$. For a stereo feature, stack $\mathbf{H}_i^L$ and $\mathbf{H}_i^R$ (4 rows). For a mono feature, a single $\mathbf{H}_i^c$ (2 rows).

Assembled $\mathbf{H}_k$. Stacking all points:

$$\mathbf{H}_k = \begin{pmatrix} \mathbf{H}_1 \\ \vdots \\ \mathbf{H}_{N_f} \end{pmatrix} \in \mathbb{R}^{(\sum_i \nu_i) \times (18+3N_f)} \tag{114}$$

The matrix is block-sparse: each point contributes a nonzero block only in its own columns.

2.6.5 Noise Jacobian $\mathbf{H}_\mathbf{v}$

Since the noise is additive (Equation 105):

$$\mathbf{H}_\mathbf{v} = \frac{\partial h}{\partial \mathbf{v}}\bigg|_{\hat{\mathcal{X}}_k^-,\, \mathbf{v}=\mathbf{0}} = \mathbf{I}_{\sum_i \nu_i} \tag{115}$$

so the gain equation (Section 2.2.1) simplifies to:

$$\mathbf{K}_k = \mathbf{P}_k^- \, \mathbf{H}_k^\top \left[\mathbf{H}_k \, \mathbf{P}_k^- \, \mathbf{H}_k^\top + \mathbf{R}\right]^{-1} \tag{116}$$

2.6.6 Indirect Correction of Non-Observed States

The measurement Jacobian has zeros in the $\delta\boldsymbol{\xi}$, $\delta\mathbf{v}$, $\delta\boldsymbol{\omega}$, $\delta\mathbf{g}^B$, and $\delta\mathbf{d}^B$ columns: the measurement model projects body-frame feature positions through fixed camera extrinsics, so pose, velocity, rotation rate, gravity, and disturbance do not appear. These states are corrected indirectly through covariance cross-correlations built during propagation: $\dot{\mathbf{p}}_i^B$ depends on $\mathbf{v}$ and $\boldsymbol{\omega}$ (Equation 79), and $\dot{T}$ depends on $\mathbf{v}$ and $\boldsymbol{\omega}$ (Equation 69), so $\mathbf{P}$ develops off-diagonal blocks that couple feature-point innovations to velocity, rotation, pose, gravity, and disturbance corrections.

2.6.7 Gravity Pseudo-Measurement

The gravity magnitude $\|\mathbf{g}^B\| = g$ is known a priori. We encode this as a scalar measurement to prevent magnitude drift:

$$\tilde{\mathbf{y}}_g = h_g(\mathcal{X}, v_g) = \|\mathbf{g}^B\|^2 + v_g = g^2 + v_g \tag{117}$$

where $v_g \sim \mathcal{N}(0, R_g)$ is scalar measurement noise. The "observation" is the known constant $\tilde{\mathbf{y}}_g = g^2$. R_g should reflect the minor variations in gravity ($g = 9.81$ is a common approximation which varies along the earths surface and decays with height). R_g should be small, unmodeled forces ought to be reflected in $\mathbf{d}^B$

Jacobians. The state Jacobian $H_g = \frac{Dh_g}{D\mathcal{X}}\Big|_{\hat{\mathcal{X}}_k,\, v_g=0}$. Since h_g depends only on $\mathbf{g}^B \in \mathbb{R}^3$ (Euclidean), the composite derivative reduces to the standard partial derivative:

$$\frac{\partial \, \|\mathbf{g}^B\|^2}{\partial \, \mathbf{g}^B} = 2(\mathbf{g}^B)^\top \tag{118}$$

All other state components do not appear in h_g, giving:

$$H_g = \Big(\mathbf{0}_{1\times 6} \quad \mathbf{0}_{1\times 3} \quad \mathbf{0}_{1\times 3} \quad 2(\hat{\mathbf{g}}^B)^\top \quad \mathbf{0}_{1\times 3} \quad \mathbf{0}_{1\times 3} \quad \cdots \quad \mathbf{0}_{1\times 3}\Big) \in \mathbb{R}^{1\times(18+3N_f)} \tag{119}$$

The noise Jacobian is:

$$H_{v_g} = \frac{\partial h_g}{\partial v_g}\bigg|_{v_g=0} = 1 \tag{120}$$

so the gain equation gives:

$$K_g = \mathbf{P}\, H_g^\top \left[H_g \, \mathbf{P}\, H_g^\top + R_g\right]^{-1} \in \mathbb{R}^{(18+3N_f)\times 1} \tag{121}$$

which is a scalar division (the bracketed term is 1×1).

Sequential update. The gravity pseudo-measurement can be processed as a separate update step after the feature-point update. This is valid because the Kalman update equations can be applied any number of times within a single time step by splitting the measurement vector into independent groups (Wescott, 2022). Each group updates $\hat{\mathcal{X}}$ and $\mathbf{P}$ using the posterior from the previous group as the prior for the next. The result is algebraically equivalent to processing all measurements simultaneously in a single large update, provided the measurement noises are independent. This is not to be confused with the iterated EKF (IEKF), which re-linearises the measurement model at the updated state and solves an optimisation problem iteratively.
We stack all measurements into one update however since deriving the pose change and it's covariance becomes easier.

2.6.8 Stationarity Assumption

The dynamics $\dot{\mathbf{p}}_i^B = -\mathbf{v} - \boldsymbol{\omega} \times \mathbf{p}_i^B$ (Equation 79) assume the point is fixed in the world. If the point is moving, the propagated position is wrong, the predicted pixel position is wrong, and the innovation is large. The chi-squared test (Section 2.11) detects this and excludes the point from the update.

2.7 Computer Vision Primitives

2.7.1 Feature Detection: FAST

FAST (Features from Accelerated Segment Test) (Rosten and Drummond, 2006) examines a circle of 16 pixels around each candidate. If a contiguous arc of N pixels (typically 9 or 12) are all brighter or darker than the centre by a threshold, it is a corner.

2.7.2 Feature Quality: Shi-Tomasi

The Shi-Tomasi cornerness score (Shi and Tomasi, 1994) is $\min(\lambda_1, \lambda_2)$ where λ_1, λ_2 are eigenvalues of the structure tensor $\mathbf{M} = \sum_{\text{patch}} \nabla I \, \nabla I^\top$. A high score means strong gradients in two independent directions — the patch is uniquely localised and suitable for tracking.

A minimum score τ_{min} rejects unreliable detections. In featureless or overexposed scenes, the pool may be empty, this is the correct behaviour.

2.7.3 Temporal Tracking: Lucas-Kanade

LK (Lucas and Kanade, 1981) (pyramidal implementation (Bouguet, 2000)) finds the displacement $\delta\mathbf{u}$ that minimises $\sum_{\text{patch}}[I_k(\mathbf{u} + \delta\mathbf{u}) - I_{k-1}(\mathbf{u})]^2$. Linearising gives:

$$\mathbf{M}\,\delta\mathbf{u} = -\sum_{\text{patch}} (I_k - I_{k-1})\,\nabla I_k \tag{122}$$

where $\mathbf{M}$ is the same structure tensor. The system is well-conditioned when both eigenvalues are large (Shi-Tomasi score high). Pyramidal LK handles large displacements by solving coarse-to-fine.

Forward-backward check. Track forward $k{-}1 \to k$ to get $\mathbf{u}'$, then backward $k \to k{-}1$ from $\mathbf{u}'$ to get $\mathbf{u}''$. If $\|\mathbf{u}'' - \mathbf{u}\| > \epsilon_{\text{fb}}$, the match is inconsistent and the point is dropped. This is the primary failure detector.

2.7.4 Stereo Matching: NCC

Normalised Cross-Correlation (Szeliski, 2022) matches a patch from one image along the epipolar line in the other:

$$\text{NCC}(\mathbf{u}_L, \mathbf{u}_R) = \frac{\sum(I_L - \bar{I}_L)(I_R - \bar{I}_R)}{\sqrt{\sum(I_L - \bar{I}_L)^2 \, \sum(I_R - \bar{I}_R)^2}} \tag{123}$$

Invariant to brightness and contrast differences between cameras. For rectified stereo the epipolar line is a horizontal row, so the search reduces to 1D over the disparity range $[d_{\text{min}}, d_{\text{max}}]$. The location of the highest NCC value, provided it exceeds threshold NCC_{min}, is the match. Sub-pixel refinement by parabolic fitting on the three response values around the peak:

$$\Delta u = \frac{1}{2}\frac{y_{-1} - y_{+1}}{y_{-1} - 2y_0 + y_{+1}}, \qquad u^* = u_{\text{peak}} + \Delta u \tag{124}$$

clamped to $[-1, 1]$. Boundary peaks skip refinement.

2.7.5 Grid Selection and Focus Weighting

The image is divided into a cols $\times$ rows grid. Both selection routines first filter the candidate pool against the pixels of currently-tracked points, discarding any candidate closer than $d_{\text{min}}^{\text{px}}$ to an existing one.

Feature point allocation. The top $N_{\text{feat},gl}$ candidates by Shi-Tomasi cornerness are picked globally. The remaining candidates are binned into the grid, and the top $N_{\text{feat},lo}$ per cell are added. The grid is purely a spatial spread mechanism and need not match the interest-point grid.

Feature replenishment ought to be weighted by expected remaining cell lifetime (from estimated velocity and angular velocity): cells the drone is moving toward get more features. This also avoids spending compute on points that fall out of the image at the next time step anyway. Feature points should remain in frame for at least 3 frames, so 50ms at 60Hz frame rate. Ideally more.

Interest point allocation. The focus point $\mathbf{F}_{k-1} \in \mathbb{R}^3$ projects into each camera as $\mathbf{f}_c = \boldsymbol{\pi}(\mathbf{R}_{cB}\,\mathbf{F}_{k-1} + \mathbf{t}_c)$, $c \in \{L, R\}$. A 2D isotropic Gaussian $\mathcal{N}(\mathbf{f}_c,\, \sigma_F^2\mathbf{I})$ defined over the image plane assigns a probability mass to each grid cell. Because the two axes are independent, the joint mass factors:

$$P(\text{cell}_{r,c}) = \frac{P(x \in [x_c, x_{c+1}])\, P(y \in [y_r, y_{r+1}])}{Z_x \cdot Z_y} \tag{125}$$

where each one-dimensional probability is a difference of normal CDFs, and $Z_x = P(0 \leq x \leq W)$, $Z_y = P(0 \leq y \leq H)$ re-normalise the distribution over the image rectangle.
Each camera receives $N_{\mathcal{I}}^{\max}/2$ points, allocated $\lfloor (N_{\mathcal{I}}^{\max}/2) \cdot P(\text{cell}_{r,c}) \rfloor$ per cell, with the top candidates per cell by Shi-Tomasi cornerness chosen. Any residual from the floor operation is distributed in weight-descending order, peak cell first, then outward. Cells that cannot fulfil their allocation (too few candidates) overflow their residual into the same descending order. Smaller σ_F concentrates points near the focus; larger σ_F spreads them. NCC stereo matching then attempts to pair each interest point across cameras; successful matches contribute stereo interest points, failures contribute mono.

2.8 Nonlinear MAP Estimation with Gauss-Newton

Following the formulation in (Haavardsholm, 2026b).

2.8.1 Measurement Model

For measurements $\mathbf{z}_i$ and latent state $\mathbf{x}$, assume:

$$\mathbf{z}_i = h_i(\mathbf{x}) + \boldsymbol{\eta}_i, \qquad \boldsymbol{\eta}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_i) \tag{126}$$

The predicted measurement is $\hat{\mathbf{z}}_i = h_i(\mathbf{x})$, and the measurement residual is:

$$e_i(\mathbf{x}) = h_i(\mathbf{x}) - \mathbf{z}_i \tag{127}$$

2.8.2 Nonlinear MAP as Weighted Least Squares

Under Gaussian noise, the MAP estimate is obtained by solving the nonlinear least squares problem:

$$\mathbf{x}^* = \arg\min_{\mathbf{x}} \sum_{i=1}^{n} \|h_i(\mathbf{x}) - \mathbf{z}_i\|^2_{\boldsymbol{\Sigma}_i} \tag{128}$$

where the squared Mahalanobis norm is:

$$\|\mathbf{e}\|^2_{\boldsymbol{\Sigma}} = \mathbf{e}^\top \boldsymbol{\Sigma}^{-1} \mathbf{e} = \|\boldsymbol{\Sigma}^{-1/2} \mathbf{e}\|^2 \tag{129}$$

This yields the whitened Jacobian and residual:

$$A_i = \boldsymbol{\Sigma}_i^{-1/2} J^{h_i}_{\mathbf{x}}\Big|_{\hat{\mathbf{x}}^t} \tag{130}$$

$$b_i = \boldsymbol{\Sigma}_i^{-1/2} (\mathbf{z}_i - h_i(\hat{\mathbf{x}}^t)) \tag{131}$$

Stack all measurements:

$$A = \begin{pmatrix} A_1 \\ \vdots \\ A_n \end{pmatrix}, \qquad b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \tag{132}$$

2.8.3 Gauss-Newton Iteration

At iteration t, linearise the residual around the current estimate $\hat{\mathbf{x}}^t$:

$$h_i(\mathbf{x}) \approx h_i(\hat{\mathbf{x}}^t) + J^{h_i}_{\mathbf{x}}\Big|_{\hat{\mathbf{x}}^t} \boldsymbol{\tau} \tag{133}$$

where $\boldsymbol{\tau}$ is the increment. The Gauss-Newton step solves:

$$\boldsymbol{\tau}^* = \arg\min_{\boldsymbol{\tau}} \|A\boldsymbol{\tau} - b\|^2 \tag{134}$$

via the normal equations $A^\top A \,\boldsymbol{\tau}^* = A^\top b$. Update:

$$\hat{\mathbf{x}}^{t+1} = \hat{\mathbf{x}}^t \oplus \boldsymbol{\tau}^* \tag{135}$$

where $\oplus$ is addition in Euclidean space or the right plus operator on the appropriate manifold. Note the Jacobian must also be defined on the manifold.

2.8.4 Estimated Covariance

The covariance of the MAP estimate is approximated by the inverse of the approximate Hessian at the solution:

$$\boldsymbol{\Sigma}_{\hat{\mathbf{x}}} \approx (A^\top A)^{-1} \tag{136}$$

so that $\mathbf{x} \approx \mathcal{N}(\hat{\mathbf{x}}, \boldsymbol{\Sigma}_{\hat{\mathbf{x}}})$.

2.8.5 Including a Prior

If a prior on part or all of the state is available, it enters as an additional term in the objective. This is seen by noticing the prior adds a quadratic penalty to the negative log-posterior ((Wilkinson et al., 2022), Equation 34). The prior is then treated as a virtual measurement and stacked into A and b alongside the measurement terms.

Full prior on Euclidean state. If $\mathbf{x} \in \mathbb{R}^n$ with prior $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$:

$$A_0 = \boldsymbol{\Sigma}_0^{-1/2}\,\mathbf{I}_n, \qquad b_0 = \boldsymbol{\Sigma}_0^{-1/2}\,(\boldsymbol{\mu}_0 - \hat{\mathbf{x}}^t) \tag{137}$$

The residual b_0 is *prior minus estimate*, consistent with the general formula $b_i = \boldsymbol{\Sigma}_i^{-1/2}(\mathbf{z}_i - h_i(\hat{\mathbf{x}}^t))$.

Partial prior. If the prior applies only to a subset of the state (e.g. $\mathbf{x} = (\mathbf{x}_a, \mathbf{x}_b)$ with prior on $\mathbf{x}_a$ only), the selection matrix replaces the identity:

$$A_0 = \boldsymbol{\Sigma}_0^{-1/2}\begin{pmatrix}\mathbf{I}_{n_a} & \mathbf{0}_{n_a \times n_b}\end{pmatrix}, \qquad b_0 = \boldsymbol{\Sigma}_0^{-1/2}\,(\boldsymbol{\mu}_0 - \hat{\mathbf{x}}_a^t) \tag{138}$$

where n_a is the dimension of $\mathbf{x}_a$. This penalises only the $\boldsymbol{\tau}_a$ increment, leaving $\boldsymbol{\tau}_b$ unconstrained by the prior.

Prior on a manifold state. If the state (or the prior subset) includes a component $\mathcal{X} \in \mathcal{M}$ on a Lie group with prior mean $\bar{\mathcal{X}}$ and covariance $\boldsymbol{\Sigma}_0$ defined on the tangent space, the Euclidean difference $\boldsymbol{\mu}_0 - \hat{\mathbf{x}}^t$ is replaced by the manifold $\ominus$ operator (Section 2.1):

$$b_0 = \boldsymbol{\Sigma}_0^{-1/2}\,(\bar{\mathcal{X}} \ominus \hat{\mathcal{X}}^t) = \boldsymbol{\Sigma}_0^{-1/2}\,\mathrm{Log}\Big((\hat{\mathcal{X}}^t)^{-1} \cdot \bar{\mathcal{X}}\Big) \tag{139}$$

This is *prior* $\ominus$ *estimate*, the manifold analogue of prior minus estimate. At $\hat{\mathcal{X}}^t = \bar{\mathcal{X}}$ the residual is $\mathrm{Log}(\mathcal{E}) = \mathbf{0}$, as required. The sign follows from the virtual measurement formulation: the residual $e_0 = \mathcal{X} \ominus \bar{\mathcal{X}} = \mathrm{Log}(\bar{\mathcal{X}}^{-1} \cdot \mathcal{X})$ and $b_0 = -\boldsymbol{\Sigma}_0^{-1/2}\, e_0|_{\hat{\mathcal{X}}^t}$, using $-\mathrm{Log}(X) = \mathrm{Log}(X^{-1})$ ((Solà et al., 2021), Eq. 19,23,24). The A_0 matrix is unchanged:

$$A_0 = \boldsymbol{\Sigma}_0^{-1/2}\,\mathbf{I}_m \tag{140}$$

where $m = \dim(\mathcal{M})$ is the tangent space dimension (e.g. $m = 6$ for $SE(3)$), padded with zeros if the prior is partial.

Combined manifold and Euclidean prior. If the prior covers both a Lie group component $\mathcal{X} \in \mathcal{M}$ and a Euclidean component $\mathbf{x}_e \in \mathbb{R}^n$ (e.g. pose and a point position), with joint prior mean $(\bar{\mathcal{X}},\ \boldsymbol{\mu}_e)$ and joint covariance $\boldsymbol{\Sigma}_0 \in \mathbb{R}^{(m+n)\times(m+n)}$, the residual is assembled block-wise using the appropriate operator for each component:

$$b_0 = \boldsymbol{\Sigma}_0^{-1/2} \begin{pmatrix} \bar{\mathcal{X}} \ominus \hat{\mathcal{X}}^t \\ \boldsymbol{\mu}_e - \hat{\mathbf{x}}_e^t \end{pmatrix} \tag{141}$$

with $A_0 = \boldsymbol{\Sigma}_0^{-1/2}(\mathbf{I}_{m+n},\ \mathbf{0})$ if the prior is partial, or $A_0 = \boldsymbol{\Sigma}_0^{-1/2}\mathbf{I}_{m+n}$ if it covers the full state. The joint covariance $\boldsymbol{\Sigma}_0$ may have nonzero off-diagonal blocks encoding correlations between the manifold and Euclidean components; the whitening $\boldsymbol{\Sigma}_0^{-1/2}$ handles this automatically.

2.8.6 Relative Pose from Filter State

The relative pose $\Delta T = T_{B_k,B_{k-1}} \in SE(3)$ and its covariance are extracted directly from the filter state.

Relative Pose After propagation at step k, the discrete estimated poses are $\hat{T}^+_{k-1}$ and $\hat{T}^-_k$ (or $\hat{T}^+_k$ after the update). From $\Delta T = T_{B_k,B_0} \cdot T^{-1}_{B_{k-1},B_0}$:

$$\Delta\hat{T} = \hat{T}_k \cdot \hat{T}^{-1}_{k-1} \tag{142}$$

with rotation and translation blocks:

$$\Delta\hat{\mathbf{R}} = \hat{\mathbf{R}}_{B_k}\,\hat{\mathbf{R}}^\top_{B_{k-1}}, \qquad \Delta\hat{\mathbf{t}} = \hat{\mathbf{t}}_{B_k} - \Delta\hat{\mathbf{R}}\,\hat{\mathbf{t}}_{B_{k-1}} \tag{143}$$

Derivation. From $\mathbf{p}^{B_k} = \mathbf{R}_{B_k}\mathbf{p}^{B_0} + \mathbf{t}_{B_k}$ and $\mathbf{p}^{B_0} = \mathbf{R}^\top_{B_{k-1}}(\mathbf{p}^{B_{k-1}} - \mathbf{t}_{B_{k-1}})$, substitute:

$$\mathbf{p}^{B_k} = \underbrace{\mathbf{R}_{B_k}\mathbf{R}^\top_{B_{k-1}}}_{\Delta\mathbf{R}}\,\mathbf{p}^{B_{k-1}} + \underbrace{\mathbf{t}_{B_k} - \mathbf{R}_{B_k}\mathbf{R}^\top_{B_{k-1}}\mathbf{t}_{B_{k-1}}}_{\Delta\mathbf{t}} \tag{144}$$

Covariance Preliminaries The following standard identities are used throughout this section. For zero-mean random vectors $\mathbf{a}$, $\mathbf{b}$ and deterministic matrices $\mathbf{M}$, $\mathbf{N}$ (Bar-Shalom et al., 2001):

Cross-covariance.

$$\mathrm{Cov}(\mathbf{a},\,\mathbf{b}) = \mathrm{E}[\mathbf{a}\,\mathbf{b}^\top] \tag{145}$$

Linear transformation. If $\mathbf{c} = \mathbf{M}\mathbf{a}$, then:

$$\mathrm{Cov}(\mathbf{c}) = \mathbf{M}\,\mathrm{Cov}(\mathbf{a})\,\mathbf{M}^\top \tag{146}$$

Cross-covariance under linear transformation. If $\mathbf{c} = \mathbf{M}\mathbf{a}$, then:

$$\mathrm{Cov}(\mathbf{c},\,\mathbf{b}) = \mathbf{M}\,\mathrm{Cov}(\mathbf{a},\,\mathbf{b}) \tag{147}$$

Independence. If $\mathbf{a}$ and $\mathbf{b}$ are independent:

$$\mathrm{Cov}(\mathbf{a},\,\mathbf{b}) = \mathbf{0} \tag{148}$$

Sum with independent term. If $\mathbf{c} = \mathbf{M}\mathbf{a} + \mathbf{n}$ with $\mathbf{n}$ independent of both $\mathbf{a}$ and $\mathbf{b}$:

$$\mathrm{Cov}(\mathbf{c},\,\mathbf{b}) = \mathbf{M}\,\mathrm{Cov}(\mathbf{a},\,\mathbf{b}) \tag{149}$$

Perturbation of the Relative Pose The filter uses right perturbation (Section 2.1.1): $T_{B_j,\text{true}} = \hat{T}_{B_j} \cdot \mathrm{Exp}(\delta\boldsymbol{\xi}_j)$ where $\delta\boldsymbol{\xi}_j \in \mathbb{R}^6$ lives in B_0 (the domain frame of T_{B_j,B_0}) for both $j = k-1$ and $j = k$. The relative pose perturbation $\delta\Delta\boldsymbol{\xi} \in \mathbb{R}^6$, defined by $\Delta T_{\text{true}} = \Delta\hat{T} \cdot \mathrm{Exp}(\delta\Delta\boldsymbol{\xi})$, lives in B_{k-1} (the domain frame of $\Delta T_{B_k,B_{k-1}}$).

Derivation. Perturb both poses:

$$\Delta T_{\text{true}} = \hat{T}_{B_k}\,\mathrm{Exp}(\delta\boldsymbol{\xi}_k) \cdot \Big(\hat{T}_{B_{k-1}}\,\mathrm{Exp}(\delta\boldsymbol{\xi}_{k-1})\Big)^{-1} \tag{150}$$

Invert the second factor:

$$= \hat{T}_{B_k}\,\mathrm{Exp}(\delta\boldsymbol{\xi}_k)\,\mathrm{Exp}(-\delta\boldsymbol{\xi}_{k-1})\,\hat{T}_{B_{k-1}}^{-1} \tag{151}$$

Both $\delta\boldsymbol{\xi}_k$ and $\delta\boldsymbol{\xi}_{k-1}$ are small filter perturbations. Apply ((Solà et al., 2021), Eq. 69):

$$\mathrm{Exp}(\boldsymbol{\tau})\,\mathrm{Exp}(\delta\boldsymbol{\tau}) \approx \mathrm{Exp}(\boldsymbol{\tau} + \mathbf{J}_r^{-1}(\boldsymbol{\tau})\,\delta\boldsymbol{\tau}) \tag{152}$$

with $\boldsymbol{\tau} = \delta\boldsymbol{\xi}_k$ and $\delta\boldsymbol{\tau} = -\delta\boldsymbol{\xi}_{k-1}$. Since $\delta\boldsymbol{\xi}_k$ is small, $\mathbf{J}_r^{-1}(\delta\boldsymbol{\xi}_k) \approx \mathbf{I}_6$: the $SE(3)$ right Jacobian evaluated at the origin gives $\mathbf{I}_6$, as $\mathbf{J}_l(\boldsymbol{\theta})$ ((Solà et al., 2021), Eq. 145) has $\sin\theta/\theta \to 1$ and $[\mathbf{u}]_\times$ terms vanishing, $Q(\boldsymbol{\rho}, \boldsymbol{\theta})$ ((Solà et al., 2021), Eq. 180) has every term proportional to $\boldsymbol{\rho}_\times$ or $\boldsymbol{\theta}_\times$ so $Q(\mathbf{0}, \mathbf{0}) = \mathbf{0}$, giving $\mathbf{J}_l(\mathbf{0}) = \mathbf{I}_6$ ((Solà et al., 2021), Eq. 179a), and $\mathbf{J}_r(\boldsymbol{\tau}) = \mathbf{J}_l(-\boldsymbol{\tau})$ ((Solà et al., 2021), Eq. 76). Therefore:

$$\Delta T_{\text{true}} = \hat{T}_{B_k}\,\mathrm{Exp}(\delta\boldsymbol{\xi}_k - \delta\boldsymbol{\xi}_{k-1})\,\hat{T}_{B_{k-1}}^{-1} \tag{153}$$

Insert $\mathcal{E} = \hat{T}_{B_{k-1}}^{-1}\hat{T}_{B_{k-1}}$ between $\hat{T}_{B_k}$ and $\mathrm{Exp}(\cdot)$:

$$= \underbrace{\hat{T}_{B_k}\,\hat{T}_{B_{k-1}}^{-1}}_{\Delta\hat{T}} \cdot \hat{T}_{B_{k-1}}\,\mathrm{Exp}(\delta\boldsymbol{\xi}_k - \delta\boldsymbol{\xi}_{k-1})\,\hat{T}_{B_{k-1}}^{-1} \tag{154}$$

Apply the adjoint property $\mathcal{X}\,\mathrm{Exp}(\boldsymbol{\tau})\,\mathcal{X}^{-1} = \mathrm{Exp}(\mathrm{Ad}_{\mathcal{X}}\,\boldsymbol{\tau})$, obtained from ((Solà et al., 2021), Eq. 32): $\mathcal{X} \oplus \boldsymbol{\tau} = (\mathrm{Ad}_{\mathcal{X}}\,\boldsymbol{\tau}) \oplus \mathcal{X}$, expanded as $\mathcal{X} \cdot \mathrm{Exp}(\boldsymbol{\tau}) = \mathrm{Exp}(\mathrm{Ad}_{\mathcal{X}}\,\boldsymbol{\tau}) \cdot \mathcal{X}$ and right-multiplied by $\mathcal{X}^{-1}$. With $\mathcal{X} = \hat{T}_{B_{k-1}}$:

$$\Delta T_{\text{true}} = \Delta\hat{T} \cdot \mathrm{Exp}\Big(\mathrm{Ad}_{\hat{T}_{B_{k-1}}}(\delta\boldsymbol{\xi}_k - \delta\boldsymbol{\xi}_{k-1})\Big) \tag{155}$$

Reading off the right perturbation of ΔT:

$$\boxed{\delta\Delta\boldsymbol{\xi} = \mathrm{Ad}_{\hat{T}_{B_{k-1}}}(\delta\boldsymbol{\xi}_k - \delta\boldsymbol{\xi}_{k-1})} \tag{156}$$

where $\mathrm{Ad}_{\hat{T}_{B_{k-1}}}$ (Equation 20) maps from B_0 (where $\delta\boldsymbol{\xi}_j$ live) to B_{k-1} (where $\delta\Delta\boldsymbol{\xi}$ lives).

Verification. At $\hat{T}_{B_{k-1}} = \mathcal{E}$: $\mathrm{Ad}_{\mathcal{E}} = \mathbf{I}_6$, so $\delta\Delta\boldsymbol{\xi} = \delta\boldsymbol{\xi}_k - \delta\boldsymbol{\xi}_{k-1}$. ✓

Jacobian and Covariance From Equation 156, the perturbation is linear in $(\delta\boldsymbol{\xi}_{k-1},\, \delta\boldsymbol{\xi}_k)$:

$$\delta\Delta\boldsymbol{\xi} = \underbrace{\begin{pmatrix} -\mathrm{Ad}_{\hat{T}_{B_{k-1}}} & \mathrm{Ad}_{\hat{T}_{B_{k-1}}} \end{pmatrix}}_{\mathbf{J}_\Delta \in \mathbb{R}^{6\times 12}} \begin{pmatrix} \delta\boldsymbol{\xi}_{k-1} \\ \delta\boldsymbol{\xi}_k \end{pmatrix} \tag{157}$$

By Equation 146:

$$\boldsymbol{\Sigma}_{\Delta\xi} = \mathbf{J}_\Delta\, \mathbf{P}_{\text{pose,joint}}\, \mathbf{J}_\Delta^\top \in \mathbb{R}^{6\times 6} \tag{158}$$

where $\mathbf{P}_{\text{pose,joint}}$ is the 12×12 joint covariance of $(\delta\boldsymbol{\xi}_{k-1},\, \delta\boldsymbol{\xi}_k)$, derived below.

Cross-covariance between time steps. The filter covariance $\mathbf{P}$ at any single time step contains only the perturbation at that time. To assemble $\mathbf{P}_{\text{pose,joint}}$, we need the cross-covariance between the perturbations at $k-1$ and k.

The discrete covariance propagation $\mathbf{P}_k^- = \Phi_{k-1}\, \mathbf{P}_{k-1}^+\, \Phi_{k-1}^\top + \Delta t_{k-1}\, G_{k-1}\, \mathcal{Q}_{k-1}\, G_{k-1}^\top$ (Section 2.2.1) corresponds to the linearised error relation:

$$\delta\mathbf{x}_k^- = \Phi_{k-1}\, \delta\mathbf{x}_{k-1}^+ + \mathbf{n}_{k-1} \tag{159}$$

where $\Phi_{k-1} = \mathbf{I} + \Delta t_{k-1}\, F_{k-1}$ is the first-order state transition matrix (the discretisation of the continuous error dynamics $\dot{\delta\mathbf{x}} = F\,\delta\mathbf{x}$, Section 2.5.7), and $\mathbf{n}_{k-1}$ is the discretised process noise with covariance $\Delta t_{k-1}\, G_{k-1}\, \mathcal{Q}_{k-1}\, G_{k-1}^\top$, independent of $\delta\mathbf{x}_{k-1}^+$.
Verification: applying Equation 146 and Equation 148:

$$\begin{aligned} \mathrm{Cov}(\delta\mathbf{x}_k^-) &= \Phi_{k-1}\, \mathrm{Cov}(\delta\mathbf{x}_{k-1}^+)\, \Phi_{k-1}^\top + \mathrm{Cov}(\mathbf{n}_{k-1}) \\ &= \Phi_{k-1}\, \mathbf{P}_{k-1}^+\, \Phi_{k-1}^\top + \Delta t_{k-1}\, G_{k-1}\, \mathcal{Q}_{k-1}\, G_{k-1}^\top = \mathbf{P}_k^- \end{aligned} \tag{160}$$

which recovers the propagation equation. ✓

The cross-covariance follows from Equation 149 with $\mathbf{c} = \delta\mathbf{x}_k^-$, $\mathbf{M} = \Phi_{k-1}$, $\mathbf{a} = \delta\mathbf{x}_{k-1}^+$, $\mathbf{b} = \delta\mathbf{x}_{k-1}^+$, and $\mathbf{n} = \mathbf{n}_{k-1}$ independent of $\delta\mathbf{x}_{k-1}^+$:

$$\mathrm{Cov}(\delta\mathbf{x}_k^-,\ \delta\mathbf{x}_{k-1}^+) = \Phi_{k-1}\, \underbrace{\mathrm{Cov}(\delta\mathbf{x}_{k-1}^+,\ \delta\mathbf{x}_{k-1}^+)}_{\mathbf{P}_{k-1}^+} = \Phi_{k-1}\, \mathbf{P}_{k-1}^+ \tag{161}$$

Pre-update joint covariance. Using $\hat{T}_k^-$ (after propagation, before update at k). Let subscript ξ denote the 6×6 pose block:

$$\mathbf{P}_{\text{pose,joint}}^- = \begin{pmatrix} \mathbf{P}_{\xi\xi}^{k-1,+} & \left(\Phi_{k-1}\, \mathbf{P}^{k-1,+}\right)_{\xi\xi}^\top \\ \left(\Phi_{k-1}\, \mathbf{P}^{k-1,+}\right)_{\xi\xi} & \mathbf{P}_{\xi\xi}^{k,-} \end{pmatrix} \tag{162}$$

where:

- $\mathbf{P}_{\xi\xi}^{k-1,+}$: the 6×6 pose block after the update at $k-1$.
- $\mathbf{P}_{\xi\xi}^{k,-}$: the 6×6 pose block after propagation to k. Larger than $\mathbf{P}_{\xi\xi}^{k-1,+}$ because propagation adds process noise.

- $(\Phi_{k-1}\,\mathbf{P}^{k-1,+})_{\xi\xi}$: pose rows of $\Phi_{k-1}\,\mathbf{P}^{k-1,+}$ restricted to pose columns. This is *not* simply $(\Phi_{k-1})_{\xi\xi}\,\mathbf{P}^{k-1,+}_{\xi\xi}$: the full product is needed because Φ_{k-1} couples pose to velocity and rotation through $F_{\delta\boldsymbol{\xi},(\delta\mathbf{v},\delta\boldsymbol{\omega})} = -\mathrm{Ad}_{\hat{T}^{-1}}$ (Equation 96), so the cross-covariance picks up contributions from the velocity and rotation uncertainty at $k-1$.

Post-update joint covariance. Using $\hat{T}_k^+$ (after the update at k). The update gives $\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-$, corresponding to the error transformation:

$$\delta\mathbf{x}_k^+ = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\,\delta\mathbf{x}_k^- + \mathbf{K}_k\mathbf{v}_k \tag{163}$$

where $\mathbf{v}_k$ is measurement noise, independent of $\delta\mathbf{x}_{k-1}^+$. *Verification*: applying Equation 146 and Equation 148 (noting $\mathrm{Cov}(\mathbf{Kv}) = \mathbf{KR}_k\mathbf{K}^\top$ and $(\mathbf{I}-\mathbf{KH})\mathbf{P}^-(\mathbf{I}-\mathbf{KH})^\top + \mathbf{KR}_k\mathbf{K}^\top = (\mathbf{I}-\mathbf{KH})\mathbf{P}^-$ by the Joseph form identity when $\mathbf{K}$ is the optimal gain):

$$\mathrm{Cov}(\delta\mathbf{x}_k^+) = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\,\mathbf{P}_k^- = \mathbf{P}_k^+ \quad \checkmark \tag{164}$$

The cross-covariance with $\delta\mathbf{x}_{k-1}^+$ follows from Equation 149 with $\mathbf{c} = \delta\mathbf{x}_k^+$, $\mathbf{M} = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)$, $\mathbf{a} = \delta\mathbf{x}_k^-$, $\mathbf{b} = \delta\mathbf{x}_{k-1}^+$, and $\mathbf{n} = \mathbf{K}_k\mathbf{v}_k$ independent of $\delta\mathbf{x}_{k-1}^+$:

$$\begin{aligned}\mathrm{Cov}(\delta\mathbf{x}_k^+,\ \delta\mathbf{x}_{k-1}^+) &= (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\,\mathrm{Cov}(\delta\mathbf{x}_k^-,\ \delta\mathbf{x}_{k-1}^+)\\ &= (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\,\Phi_{k-1}\,\mathbf{P}^{k-1,+}\end{aligned} \tag{165}$$

substituting Equation 161 in the second step. The 12×12 joint covariance becomes:

$$\mathbf{P}^+_{\text{pose,joint}} = \begin{pmatrix} \mathbf{P}^{k-1,+}_{\xi\xi} & \Big((\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\,\Phi_{k-1}\,\mathbf{P}^{k-1,+}\Big)^\top_{\xi\xi} \\ \Big((\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\,\Phi_{k-1}\,\mathbf{P}^{k-1,+}\Big)_{\xi\xi} & \mathbf{P}^{k,+}_{\xi\xi} \end{pmatrix} \tag{166}$$

Which version to use. The pre-update version (Equation 162) is simpler: it is available immediately after propagation and requires only $\Phi_{k-1}\,\mathbf{P}^{k-1,+}$. The post-update version (Equation 166) is tighter: the update at k reduces $\mathbf{P}^{k,+}_{\xi\xi}$ relative to $\mathbf{P}^{k,-}_{\xi\xi}$ and tightens the cross-covariance, yielding a more informative prior. Since the pose is corrected indirectly through feature cross-correlations (Section 2.6), the improvement depends on the quality of feature matches at k. In practice, the post-update version should be used when the joint solver runs after the filter update.

Implementation. Before propagation at step k, store the pose rows of $\mathbf{P}^{k-1,+}$ (a $6 \times n$ slice where $n = 18 + 3N_f$). After propagation, compute $(\Phi_{k-1}\,\mathbf{P}^{k-1,+})_{\xi\xi}$ by extracting the pose rows of the product and restricting to pose columns. For the post-update version, apply $(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)$ to this slice after the update. This is $O(n)$ additional storage per frame.

Conversion to $SE(3)$ Lie Algebra For the joint solver (Section 2.8.7), the prior is expressed in the Lie algebra:

$$\Delta\hat{\boldsymbol{\xi}} = \mathrm{Log}(\Delta\hat{T}) \in \mathbb{R}^6, \qquad \boldsymbol{\Sigma}_{\text{prior}} = \mathbf{J}_r^{-1}\,\boldsymbol{\Sigma}_{\Delta\xi}\,\mathbf{J}_r^{-\top} \tag{167}$$

where $\mathbf{J}_r^{-1}$ is evaluated at $\Delta\hat{\boldsymbol{\xi}}$ ((Solà et al., 2021), Eq. 79: $\mathbf{J}_{\mathcal{X}}^{\mathrm{Log}(\mathcal{X})} = \mathbf{J}_r^{-1}(\boldsymbol{\tau})$ with $\boldsymbol{\tau} = \mathrm{Log}(\mathcal{X})$). For one frame interval at typical frame rates, the relative pose is near the identity, $\mathbf{J}_r^{-1} \approx \mathbf{I}_6$ (same argument as above: (Solà et al., 2021), Eqs. 179a, 145, 76), and $\boldsymbol{\Sigma}_{\mathrm{prior}} \approx \boldsymbol{\Sigma}_{\Delta\xi}$.

2.8.7 Application: Joint Point-Velocity Solver with Pose Prior

The two-frame point-velocity solver estimates position and velocity for all tracked points jointly with the pose change, using the filter-derived pose (Section 2.8.6) as a prior. The structure follows the MAP framework of Section 2.8, applied to a joint state containing one $SE(3)$ pose and N Euclidean per-point states. This is a two-view bundle adjustment with two frame pairs (Haavardsholm, 2026a). An interesting observation is that for every $n'th$ frame correspondence you can estimate the $n'th$ order derivative of position.

Joint State The state to be estimated is:

$$\mathbf{x}_{\mathrm{joint}} = \langle\, \Delta T,\ \mathbf{s}_1,\ \ldots,\ \mathbf{s}_N \,\rangle \tag{168}$$

where $\Delta T = T_{B_k, B_{k-1}} \in SE(3)$ is the pose change, maintained as a group element. Per point:

- SS, SM, MS: $\mathbf{s}_i = (\mathbf{p}_i^{B_{k-1}},\ \mathbf{v}_i^{\mathbf{p}}) \in \mathbb{R}^6$
- MM: $\mathbf{s}_i = (\mathbf{p}_i^{B_{k-1}},\ v_{\perp,i}) \in \mathbb{R}^4$

The Gauss-Newton increment (Section 2.8) is:

$$\boldsymbol{\tau} = \begin{pmatrix} \boldsymbol{\tau}_\xi \\ \boldsymbol{\tau}_{s_1} \\ \vdots \\ \boldsymbol{\tau}_{s_N} \end{pmatrix} \in \mathbb{R}^{6+\sum_i \dim(\mathbf{s}_i)} \tag{169}$$

where $\boldsymbol{\tau}_\xi \in \mathbb{R}^6$ is a fresh tangent vector at each iteration. The update uses $\oplus$ appropriate to each component:

$$\Delta T^{t+1} = \Delta T^t \cdot \mathrm{Exp}(\boldsymbol{\tau}_\xi), \qquad \mathbf{s}_i^{t+1} = \mathbf{s}_i^t + \boldsymbol{\tau}_{s_i} \tag{170}$$

The $SE(3)$ update composes on the right (Section 2.1), so $\boldsymbol{\tau}_\xi$ is always a small perturbation near the identity regardless of how far ΔT is from $\mathcal{E}$.

MM is observable only when the camera center moves between frames: $\|\mathbf{t}_{\mathrm{base}}^c\| > 0$ is required for the cross product in Equation 183 to define a non-degenerate epipolar plane. Pure rotation gives $\mathbf{t}_{\mathrm{base}}^c = \mathbf{0}$ (Section 2.8.7); a numerical threshold on $\|\mathbf{t}_{\mathrm{base}}^c\|$ excludes MM points from the joint solver when this fails.

Measurement Model Given the joint state, the measurement prediction proceeds in two steps: (1) transport the point from B_{k-1} to B_k using ΔT; (2) project through the camera.

Step 1: Transport. For SS/SM/MS (full velocity), the transported point is the $SE(3)$ action $\Delta T \cdot \mathbf{p}$ with $\mathbf{p} = \mathbf{p}_i^{B_{k-1}} + \mathbf{v}_i^{\mathbf{p}}\, \Delta t_{k-1}$:

$$\mathbf{p}_i^{B_k} = \Delta T \cdot \mathbf{p} = \Delta \mathbf{R}\,(\mathbf{p}_i^{B_{k-1}} + \mathbf{v}_i^{\mathbf{p}}\, \Delta t_{k-1}) + \Delta \mathbf{t} \tag{171}$$

where $\Delta \mathbf{R}$ and $\Delta \mathbf{t}$ are the rotation and translation blocks of ΔT. For MM (perpendicular velocity only):

$$\mathbf{p}_i^{B_k} = \Delta \mathbf{R}\,(\mathbf{p}_i^{B_{k-1}} + v_{\perp,i}\, \mathbf{d}_\perp\, \Delta t_{k-1}) + \Delta \mathbf{t} \tag{172}$$

where $\mathbf{d}_\perp$ is the perpendicular direction (Section 2.8.7).

Step 2: Projection. Project into camera $c \in \{L, R\}$ (Equation 45):

$$h_{i,c}^{k} = \boldsymbol{\pi}(\mathbf{R}_{cB}\, \mathbf{p}_i^{B_k} + \mathbf{t}_c) \tag{173}$$

At $k-1$, the projection uses $\mathbf{p}_i^{B_{k-1}}$ directly (no transport):

$$h_{i,c}^{k-1} = \boldsymbol{\pi}(\mathbf{R}_{cB}\, \mathbf{p}_i^{B_{k-1}} + \mathbf{t}_c) \tag{174}$$

Assembled measurement models. Stacking the projections for each visibility type:

$$\textbf{SS: } \mathbf{h}_i = \begin{pmatrix} h_{i,L}^{k-1} \\ h_{i,R}^{k-1} \\ h_{i,L}^{k} \\ h_{i,R}^{k} \end{pmatrix} \in \mathbb{R}^8, \quad \textbf{SM: } \mathbf{h}_i = \begin{pmatrix} h_{i,L}^{k-1} \\ h_{i,R}^{k-1} \\ h_{i,c}^{k} \end{pmatrix} \in \mathbb{R}^6,$$

$$\textbf{MS: } \mathbf{h}_i = \begin{pmatrix} h_{i,c}^{k-1} \\ h_{i,L}^{k} \\ h_{i,R}^{k} \end{pmatrix} \in \mathbb{R}^6, \quad \textbf{MM: } \mathbf{h}_i = \begin{pmatrix} h_{i,c}^{k-1} \\ h_{i,c}^{k} \end{pmatrix} \in \mathbb{R}^4$$

The corresponding measurements $\mathbf{z}_i$ are the tracked pixel coordinates at both frames. Measurement noise: $\boldsymbol{\Sigma}_{z,i} = \sigma_{\text{px}}^2\, \mathbf{I}_{\nu_i}$ with ν_i matching the row count.

Measurement Jacobian Following Section 2.8, the Jacobian $J_{\mathbf{x}}^{\mathbf{h}_i}$ is the derivative of $\mathbf{h}_i$ with respect to the increment $\boldsymbol{\tau}$ (Equation 169), evaluated at $\boldsymbol{\tau} = \mathbf{0}$. For the $\mathbb{R}^3$ per-point states this is the standard partial derivative. For the $SE(3)$ pose it is the manifold Jacobian: the derivative with respect to the right perturbation $\boldsymbol{\tau}_\xi$ ((Solà et al., 2021), Eq. 41a).

For the SS case, the full Jacobian with respect to the joint state is:

$$\textbf{SS: } J_{\mathbf{x}}^{\mathbf{h}_i} = \begin{pmatrix} \frac{D\, h_{i,L}^{k-1}}{D\,\Delta T} & \frac{\partial\, h_{i,L}^{k-1}}{\partial\, \mathbf{s}_1} & \cdots & \frac{\partial\, h_{i,L}^{k-1}}{\partial\, \mathbf{s}_N} \\ \frac{D\, h_{i,R}^{k-1}}{D\,\Delta T} & \frac{\partial\, h_{i,R}^{k-1}}{\partial\, \mathbf{s}_1} & \cdots & \frac{\partial\, h_{i,R}^{k-1}}{\partial\, \mathbf{s}_N} \\ \frac{D\, h_{i,L}^{k}}{D\,\Delta T} & \frac{\partial\, h_{i,L}^{k}}{\partial\, \mathbf{s}_1} & \cdots & \frac{\partial\, h_{i,L}^{k}}{\partial\, \mathbf{s}_N} \\ \frac{D\, h_{i,R}^{k}}{D\,\Delta T} & \frac{\partial\, h_{i,R}^{k}}{\partial\, \mathbf{s}_1} & \cdots & \frac{\partial\, h_{i,R}^{k}}{\partial\, \mathbf{s}_N} \end{pmatrix}$$

where $D/D\,\Delta T$ denotes the manifold derivative with respect to the right perturbation $\boldsymbol{\tau}_\xi$ at $\boldsymbol{\tau}_\xi = \mathbf{0}$, and $\partial/\partial\,\mathbf{s}_j$ is the standard partial derivative. For other correspondence types, remove the corresponding row; for MM recall that $\mathbf{h}_i$ is a different function.

Only the transported point $\mathbf{p}_i^{B_k}$ depends on the pose change, and there is no dependence between points, so only the column corresponding to point i is nonzero:

$$\textbf{SS: } J_{\mathbf{x}}^{\mathbf{h}_i} = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,L}^{k-1}}{\partial\, \mathbf{s}_i} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,R}^{k-1}}{\partial\, \mathbf{s}_i} & \mathbf{0} & \cdots & \mathbf{0} \\ \frac{D\, h_{i,L}^{k}}{D\,\Delta T} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,L}^{k}}{\partial\, \mathbf{s}_i} & \mathbf{0} & \cdots & \mathbf{0} \\ \frac{D\, h_{i,R}^{k}}{D\,\Delta T} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,R}^{k}}{\partial\, \mathbf{s}_i} & \mathbf{0} & \cdots & \mathbf{0} \end{pmatrix}$$

We can simplify further since only the transported point depends on velocity:

$$\textbf{SS: } J_{\mathbf{x}}^{\mathbf{h}_i} = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,L}^{k-1}}{\partial\, \mathbf{p}_i^{B_{k-1}}} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,R}^{k-1}}{\partial\, \mathbf{p}_i^{B_{k-1}}} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \frac{D\, h_{i,L}^{k}}{D\,\Delta T} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,L}^{k}}{\partial\, \mathbf{p}_i^{B_{k-1}}} & \frac{\partial\, h_{i,L}^{k}}{\partial\, \mathbf{v}_i^{\mathbf{p}}} & \mathbf{0} & \cdots & \mathbf{0} \\ \frac{D\, h_{i,R}^{k}}{D\,\Delta T} & \mathbf{0} & \cdots & \mathbf{0} & \frac{\partial\, h_{i,R}^{k}}{\partial\, \mathbf{p}_i^{B_{k-1}}} & \frac{\partial\, h_{i,R}^{k}}{\partial\, \mathbf{v}_i^{\mathbf{p}}} & \mathbf{0} & \cdots & \mathbf{0} \end{pmatrix}$$

It remains to find each nonzero block. Let $\mathbf{M}_c^{k-1} = \frac{\partial \boldsymbol{\pi}}{\partial \mathbf{q}}\Big|_{k-1} \mathbf{R}_{cB}$ and $\mathbf{M}_c^{k} = \frac{\partial \boldsymbol{\pi}}{\partial \mathbf{q}}\Big|_{k} \mathbf{R}_{cB}$ denote the projection-times-extrinsic Jacobians (Section 2.6).

Per-point Jacobians w.r.t. $\mathbf{s}_i$. At $k-1$ (no dependence on velocity or pose):

$$\frac{\partial h_{i,c}^{k-1}}{\partial \mathbf{p}_i^{B_{k-1}}} = \mathbf{M}_c^{k-1}, \qquad \frac{\partial h_{i,c}^{k-1}}{\partial \mathbf{v}_i^{\mathbf{p}}} = \mathbf{0} \tag{175}$$

At k (SS/SM/MS), by the chain rule through Equation 171:

$$\frac{\partial h_{i,c}^{k}}{\partial \mathbf{p}_i^{B_{k-1}}} = \mathbf{M}_c^{k}\, \Delta\hat{\mathbf{R}}, \qquad \frac{\partial h_{i,c}^{k}}{\partial \mathbf{v}_i^{\mathbf{p}}} = \mathbf{M}_c^{k}\, \Delta\hat{\mathbf{R}}\, \Delta t_{k-1} \tag{176}$$

At k (MM), by the chain rule through Equation 172:

$$\frac{\partial h_{i,c}^{k}}{\partial \mathbf{p}_i^{B_{k-1}}} = \mathbf{M}_c^{k}\, \Delta\hat{\mathbf{R}}, \qquad \frac{\partial h_{i,c}^{k}}{\partial v_{\perp,i}} = \mathbf{M}_c^{k}\, \Delta\hat{\mathbf{R}}\, \mathbf{d}_\perp\, \Delta t_{k-1} \tag{177}$$

Jacobian w.r.t. ΔT. The measurement at k is the composition $h_{i,c}^k = \boldsymbol{\pi} \circ (\mathbf{R}_{cB}\,(\cdot) + \mathbf{t}_c) \circ (\Delta T \cdot \mathbf{p})$. By the chain rule, the manifold Jacobian with respect to ΔT is the product of the Jacobians of each stage ((Solà et al., 2021), Eq. 41):

$$J_{\Delta T}^{h_{i,c}^k} = \underbrace{\left.\frac{\partial \boldsymbol{\pi}}{\partial \mathbf{q}}\right|_k}_{\text{projection}} \cdot \underbrace{\mathbf{R}_{cB}}_{\text{extrinsics}} \cdot \underbrace{J_{\Delta T}^{\Delta T \cdot \mathbf{p}}}_{\text{action}} \tag{178}$$

The action Jacobian $J_{\Delta T}^{\Delta T \cdot \mathbf{p}}$ is the derivative of the $SE(3)$ action on $\mathbf{p}$ with respect to the right perturbation of ΔT ((Solà et al., 2021), Appendix D Eq. 182):

$$J_{\Delta T}^{\Delta T \cdot \mathbf{p}} = \begin{pmatrix} \Delta\hat{\mathbf{R}} & -\Delta\hat{\mathbf{R}}[\mathbf{p}]_\times \end{pmatrix} \in \mathbb{R}^{3\times 6} \tag{179}$$

This is defined as $\frac{\partial}{\partial \boldsymbol{\tau}_\xi}[(\Delta T \cdot \mathrm{Exp}(\boldsymbol{\tau}_\xi)) \cdot \mathbf{p}]\Big|_{\boldsymbol{\tau}_\xi = \mathbf{0}}$ — the differentiation through Exp is already performed in deriving Eq. 182; no additional chain rule is needed. Since ΔT acts directly on $\mathbf{p}$ (no inversion), there is no inversion Jacobian ((Solà et al., 2021), Eq. 178) unlike the motion-only BA formulation of (Haavardsholm, 2026a). Substituting Equation 179 into Equation 178:

$$J_{\Delta T}^{h_{i,c}^k} = \mathbf{M}_c^k \begin{pmatrix} \Delta\hat{\mathbf{R}} & -\Delta\hat{\mathbf{R}}[\mathbf{p}]_\times \end{pmatrix} \tag{180}$$

At $k{-}1$: $J_{\Delta T}^{h_{i,c}^{k-1}} = \mathbf{0}$ (the $k{-}1$ projection does not depend on the pose change).

Compact nonzero block for one SS point. Extracting the nonzero columns (ΔT and $\mathbf{s}_i$ only, 8 rows $\times$ 12 cols):

$$J_i^{\mathrm{nz}} = \begin{pmatrix} \mathbf{0}_{2\times 6} & \mathbf{M}_L^{k-1} & \mathbf{0} \\ \mathbf{0}_{2\times 6} & \mathbf{M}_R^{k-1} & \mathbf{0} \\ J_{\Delta T}^{h_{i,L}^k} & \mathbf{M}_L^k \Delta\hat{\mathbf{R}} & \mathbf{M}_L^k \Delta\hat{\mathbf{R}} \Delta t \\ J_{\Delta T}^{h_{i,R}^k} & \mathbf{M}_R^k \Delta\hat{\mathbf{R}} & \mathbf{M}_R^k \Delta\hat{\mathbf{R}} \Delta t \end{pmatrix} \tag{181}$$

SM: remove the second-camera row at k (6 measurements). MS: remove the second-camera row at $k{-}1$ (6 measurements). MM: single camera at both frames, velocity column is $\mathbf{M}_c^k \Delta\hat{\mathbf{R}} \mathbf{d}_\perp \Delta t$ (scalar, 4 measurements).

Perpendicular Direction (MM) A two-frame mono correspondence is geometrically restricted: velocity in the epipolar plane is indistinguishable from a stationary point at a different depth. Only the perpendicular component is observable.

Compute the baseline vector between camera c at $k{-}1$ and camera c at k via $T_{c_{k-1},c_k} = T_{cB} \cdot \Delta T^{-1} \cdot T_{Bc}$, using $T_{cB} = [\mathbf{R}_{cB},\, \mathbf{t}_c;\, \mathbf{0}^\top,\, 1]$, $T_{Bc} = [\mathbf{R}_{Bc},\, -\mathbf{R}_{Bc}\mathbf{t}_c;\, \mathbf{0}^\top,\, 1]$, and $\Delta T^{-1} = [\Delta\mathbf{R}^\top,\, -\Delta\mathbf{R}^\top \Delta\mathbf{t};\, \mathbf{0}^\top,\, 1]$ ((Solà et al., 2021), Appendix D Eq. 170). The transform T_{c_{k-1},c_k} maps from c_k to c_{k-1}; its translation column is the c_k origin in c_{k-1} — the baseline from c_{k-1} toward c_k, expressed in c_{k-1}. Computing the 4×4 product:

$$\mathbf{t}^c_{\text{base}} = \mathbf{t}_c - \mathbf{R}_{cB}\Delta\mathbf{R}^\top(\mathbf{R}_{Bc}\mathbf{t}_c + \Delta\mathbf{t}) \tag{182}$$

Verification: $\Delta\mathbf{R} = \mathbf{I}$, $\Delta\mathbf{t} = \mathbf{0}$ gives $\mathbf{t}^c_{\text{base}} = \mathbf{0}$. ✓

For a point with undistorted normalised coordinates (x_n, y_n) in camera c at $k-1$, the ray is $\tilde{\mathbf{x}} = (x_n,\, y_n,\, 1)^\top$. The ray and baseline both live in c_{k-1}. Their cross product gives the normal to the epipolar plane ((Opsahl and Haavardsholm, 2023), slide 15):

$$\mathbf{n}^c = \mathbf{t}^c_{\text{base}} \times \tilde{\mathbf{x}} \tag{183}$$

The perpendicular direction in body frame:

$$\mathbf{d}_\perp = \frac{\mathbf{R}_{Bc}\,\mathbf{n}^c}{\|\mathbf{n}^c\|} \tag{184}$$

Since $\mathbf{d}_\perp$ depends on ΔT (through $\Delta\mathbf{R}$ and $\Delta\mathbf{t}$ in $\mathbf{t}^c_{\text{base}}$), it is recomputed at each Gauss-Newton iteration.

Pose Prior The filter-derived relative pose $\Delta\hat{T}_{\text{EKF}}$ with covariance $\boldsymbol{\Sigma}_{\text{prior}}$ (Section 2.8.6, Equation 167) enters as a partial manifold prior on ΔT (Section 2.8.5):

$$A_0 = \boldsymbol{\Sigma}_{\text{prior}}^{-1/2}\begin{pmatrix}\mathbf{I}_6 & \mathbf{0} & \cdots\end{pmatrix}, \quad b_0 = \boldsymbol{\Sigma}_{\text{prior}}^{-1/2}\,(\Delta\hat{T}_{\text{EKF}} \ominus \Delta T^t) \tag{185}$$

where $\Delta\hat{T}_{\text{EKF}} \ominus \Delta T^t = \text{Log}((\Delta T^t)^{-1} \cdot \Delta\hat{T}_{\text{EKF}})$ is prior $\ominus$ estimate (Section 2.8.5). At $\Delta T^t = \Delta\hat{T}_{\text{EKF}}$, $b_0 = \mathbf{0}$. ✓

The $(\mathbf{I}_6,\, \mathbf{0},\, \ldots)$ selects only the $\boldsymbol{\tau}_\xi$ columns, leaving the per-point states unconstrained by the prior. The pose prior anchors the 6-DOF gauge freedom of the two-view problem, ensuring the Hessian is non-singular (Haavardsholm, 2026a).

Point Priors Per-point priors on the stereo-observed position regularise the joint solve. SS is observable without a prior, but the projection Jacobians for $\mathbf{p}$ and $\mathbf{v}^\mathbf{p}$ at frame k differ only by a factor of Δt_{k-1} (Equation 176), giving condition number $\propto 1/\Delta t_{k-1}$ on the per-point block. SM and MS drop below the unknown count when $\mathbf{v}^\mathbf{p}$ is included and are rank-deficient without further information.

The prior comes from stereo triangulation (and can come from the previous joint solve) at the frame where both cameras observe the point (Section 2.9). Stereo at frame $k-1$ (SS, SM) gives $(\hat{\mathbf{p}}_i^{B_{k-1}},\, \boldsymbol{\Sigma}_{p,i}^{B_{k-1}})$ as a direct prior on $\mathbf{p}_i^{B_{k-1}}$. Stereo at frame k (MS) gives $(\hat{\mathbf{p}}_i^{B_k},\, \boldsymbol{\Sigma}_{p,i}^{B_k})$ as a prior on the transported point $\mathbf{q}_i = \Delta T \cdot (\mathbf{p}_i^{B_{k-1}} + \mathbf{v}_i^\mathbf{p}\,\Delta t_{k-1})$ (Equation 171). In principle MM could use a previous-solve prior; currently MM is excluded entirely due to numerical instability.

Prior at $k-1$. The state component $\mathbf{p}_i^{B_{k-1}}$ is directly part of $\mathbf{s}_i$; the prior reduces to a standard partial Gaussian prior (Section 2.8.5):

$$A_{p,i} = \boldsymbol{\Sigma}_{p,i}^{-1/2} \begin{pmatrix} \mathbf{0}_{3\times 6} & \cdots & \mathbf{S}_{p,i} & \cdots \end{pmatrix}, \quad b_{p,i} = \boldsymbol{\Sigma}_{p,i}^{-1/2} \left(\hat{\mathbf{p}}_i^{B_{k-1}} - \mathbf{p}_i^{B_{k-1},t}\right) \tag{186}$$

where $\mathbf{S}_{p,i}$ selects rows 1-3 of the $\mathbf{s}_i$ columns.

Prior at k. The prior is on the transported point $\mathbf{q}_i(\mathbf{s}_i,\, \Delta T)$. Linearising at the current iterate gives the Jacobian blocks (using Equation 176 and Equation 179):

$$\frac{\partial \mathbf{q}_i}{\partial \mathbf{p}_i^{B_{k-1}}} = \Delta\hat{\mathbf{R}}, \quad \frac{\partial \mathbf{q}_i}{\partial \mathbf{v}_i^{\mathbf{p}}} = \Delta\hat{\mathbf{R}}\, \Delta t_{k-1} \tag{187}$$

$$\frac{D\mathbf{q}_i}{D\,\Delta T} = \begin{pmatrix} \Delta\hat{\mathbf{R}} & -\Delta\hat{\mathbf{R}}\,[\mathbf{p}_i + \mathbf{v}_i^{\mathbf{p}}\,\Delta t_{k-1}]_\times \end{pmatrix} \tag{188}$$

The whitened row stacks these into the pose and per-point columns:

$$A_{q,i} = \boldsymbol{\Sigma}_{p,i}^{-1/2} \begin{pmatrix} \frac{D\mathbf{q}_i}{D\,\Delta T} & \mathbf{0} & \cdots & \Delta\hat{\mathbf{R}} & \Delta\hat{\mathbf{R}}\,\Delta t_{k-1} & \cdots \end{pmatrix}, \quad b_{q,i} = \boldsymbol{\Sigma}_{p,i}^{-1/2} \left(\hat{\mathbf{p}}_i^{B_k} - \mathbf{q}_i(\mathbf{s}_i^t,\, \Delta T^t)\right) \tag{189}$$

This is a linearised manifold prior (Section 2.8.5, manifold case): A and b are rebuilt each Gauss-Newton iteration because $\mathbf{q}_i$ depends nonlinearly on both ΔT and $\mathbf{s}_i$. Previous-solve priors on points are deferred to future work; stereo triangulation is currently the only source. MM points have been excluded entirely for now due to numerical instability for which we've found no solution for yet.

Whitened System and Solution Following Section 2.8, whiten each per-point measurement block:

$$A_i = \boldsymbol{\Sigma}_{z,i}^{-1/2}\, J_{\mathbf{x}}^{\mathbf{h}_i}, \qquad b_i = \boldsymbol{\Sigma}_{z,i}^{-1/2} \left(\mathbf{z}_i - \mathbf{h}_i(\hat{\mathbf{x}}_{\text{joint}}^t)\right) \tag{190}$$

Stack the prior and all point measurements:

$$A = \begin{pmatrix} A_0 \\ A_1 \\ \vdots \\ A_N \end{pmatrix}, \quad b = \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_N \end{pmatrix} \tag{191}$$

Solve the normal equations (Section 2.8):

$$(A^\top A)\, \boldsymbol{\tau}^* = A^\top b \tag{192}$$

Update:

$$\Delta T^{t+1} = \Delta T^t \cdot \operatorname{Exp}(\boldsymbol{\tau}_\xi^*), \qquad \mathbf{s}_i^{t+1} = \mathbf{s}_i^t + \boldsymbol{\tau}_{s_i}^* \tag{193}$$

Iterate until $\|\boldsymbol{\tau}^*\| < \epsilon_1$ or $\|b\| < \epsilon_2$.

Covariance at convergence.

$$\boldsymbol{\Sigma}_{\hat{\mathbf{x}}} = (A^\top A)^{-1} \tag{194}$$

The diagonal blocks give the marginal covariances: $\boldsymbol{\Sigma}^{\text{post}}_{\Delta T}$ (6×6) for the pose, and $\boldsymbol{\Sigma}_{s_i}$ for each point $(6 \times 6$ or $4 \times 4)$.

Initialisation $\Delta T^0 = \Delta \hat{T}_{\text{EKF}}$ from Section 2.8.6. Per-point initialisation comes from the point prior or from the search step (Section 2.10): the input sample (depth, velocity) that produced the winning LK candidate provides $(\hat{\mathbf{p}}_i^0,\, \hat{\mathbf{v}}_i^{\mathbf{p},0})$.

Transform to B_k. After convergence, transform position and velocity to the current body frame. For SS/SM/MS points ($\mathbf{s}_i = (\mathbf{p}_i,\, \mathbf{v}_i^{\mathbf{p}}) \in \mathbb{R}^6$):

$$\mathbf{p}_i^{B_k} = \Delta\hat{\mathbf{R}}\,(\hat{\mathbf{p}}_i^{B_{k-1}} + \hat{\mathbf{v}}_i^{\mathbf{p}}\,\Delta t_{k-1}) + \Delta\hat{\mathbf{t}} \tag{195}$$

$$\mathbf{v}_i^{\mathbf{p},B_k} = \Delta\hat{\mathbf{R}}\,\hat{\mathbf{v}}_i^{\mathbf{p}} \tag{196}$$

The Jacobian of this transformation with respect to $\mathbf{s}_i$ is:

$$\mathbf{T}_{\Delta,i} = \begin{pmatrix} \Delta\hat{\mathbf{R}} & \Delta\hat{\mathbf{R}}\,\Delta t_{k-1} \\ \mathbf{0} & \Delta\hat{\mathbf{R}} \end{pmatrix} \in \mathbb{R}^{6\times 6} \tag{197}$$

The off-diagonal block $\Delta\hat{\mathbf{R}}\,\Delta t_{k-1}$ captures the velocity-to-position coupling from the transport. For MM points ($\mathbf{s}_i = (\mathbf{p}_i,\, v_{\perp,i}) \in \mathbb{R}^4$):

$$\mathbf{T}^{\text{MM}}_{\Delta,i} = \begin{pmatrix} \Delta\hat{\mathbf{R}} & \Delta\hat{\mathbf{R}}\,\mathbf{d}_\perp\,\Delta t_{k-1} \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix} \in \mathbb{R}^{4\times 4} \tag{198}$$

The scalar $v_\perp$ is a magnitude along $\mathbf{d}_\perp$ and does not rotate, but contributes to position uncertainty through $\mathbf{d}_\perp\,\Delta t_{k-1}$. Can also transport $\mathbf{d}_\perp$ to B_k using $\Delta\hat{R}$.

Covariance transformation. The full joint solver covariance $\boldsymbol{\Sigma}_{\hat{\mathbf{x}}} = (A^\top A)^{-1}$ (Section 2.8.7) has block structure over $(\Delta T,\, \mathbf{s}_1,\, \ldots,\, \mathbf{s}_N)$. The pose perturbation $\delta\boldsymbol{\xi}$ lives in the tangent space at ΔT (anchored in B_{k-1}) and is not transformed. The per-point states are transformed by $\mathbf{T}_{\Delta,i}$. The full block-diagonal transformation:

$$\mathbf{T} = \text{diag}(\mathbf{I}_6,\ \mathbf{T}_{\Delta,1},\ \ldots,\ \mathbf{T}_{\Delta,N}) \tag{199}$$

Applied via (Equation 146):

$$\boldsymbol{\Sigma}^{B_k} = \mathbf{T}\,\boldsymbol{\Sigma}_{\hat{\mathbf{x}}}\,\mathbf{T}^\top \tag{200}$$

This correctly leaves $\boldsymbol{\Sigma}_{\xi\xi}$ unchanged, transforms per-point blocks $\boldsymbol{\Sigma}^{B_k}_{s_i s_j} = \mathbf{T}_{\Delta,i}\,\boldsymbol{\Sigma}_{s_i s_j}\,\mathbf{T}^\top_{\Delta,j}$, and transforms the pose–point cross-covariances on the point side only: $\boldsymbol{\Sigma}^{B_k}_{\xi s_i} = \boldsymbol{\Sigma}_{\xi s_i}\,\mathbf{T}^\top_{\Delta,i}$.

2.9 Geometric Estimation

Stereo Depth From a stereo match with pixel coordinates (u_L, v_L) in the left camera and (u_R, v_R) in the right camera, first undistort to normalised coordinates:

$$(x_n, y_n) = D^{-1}\Big(\kappa^{-1}(u_L, v_L)\Big) \tag{201}$$

Compute depth from disparity and reconstruct the 3D point in the left camera frame:

$$Z = \frac{f_u\, b}{u_L - u_R}, \qquad \mathbf{p}^c = Z \begin{pmatrix} x_n \\ y_n \\ 1 \end{pmatrix} \tag{202}$$

Transform to body frame:

$$\mathbf{p}^B = \mathbf{R}_{BL}(\mathbf{p}^c - \mathbf{t}_L) \tag{203}$$

Covariance Initialisation The covariance of the reconstructed 3D position is obtained by first-order uncertainty propagation. Given a function $\mathbf{y} = g(\mathbf{x})$ with $\mathbf{x} \sim \mathcal{N}(\bar{\mathbf{x}}, \boldsymbol{\Sigma}_x)$, the output covariance is (Bar-Shalom et al., 2001):

$$\boldsymbol{\Sigma}_y = \mathbf{J}\,\boldsymbol{\Sigma}_x\,\mathbf{J}^\top, \qquad \mathbf{J} = \frac{\partial g}{\partial \mathbf{x}}\bigg|_{\bar{\mathbf{x}}} \tag{204}$$

The raw measurements are (u_L, v_L, u_R) — three independent pixel coordinates, each with noise σ_{px}. Note that u_L affects both the normalised coordinates (x_n, y_n) and the depth $Z = f_u b/(u_L - u_R)$, so these are correlated and cannot be treated independently.

We derive $\partial \mathbf{p}^c / \partial(u_L, v_L, u_R)$ using the product rule on $\mathbf{p}^c = Z \cdot (x_n, y_n, 1)^\top$.

Intermediate quantities. Depth derivatives:

$$\frac{\partial Z}{\partial u_L} = -\frac{f_u b}{(u_L - u_R)^2} = -\frac{Z}{d}, \quad \frac{\partial Z}{\partial v_L} = 0, \quad \frac{\partial Z}{\partial u_R} = \frac{Z}{d} \tag{205}$$

where $d = u_L - u_R$.

Normalised coordinate derivatives (chain rule through undistortion):

$$\frac{\partial(x_n, y_n)}{\partial(u_L, v_L)} = \underbrace{\frac{\partial D^{-1}}{\partial(x_d, y_d)}}_{\mathbf{J}_{D^{-1}}} \cdot \underbrace{\frac{\partial \kappa^{-1}}{\partial(u, v)}}_{\mathbf{K}^{-1}_{2\times 2}} \triangleq \mathbf{C} = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} \tag{206}$$

where κ^{-1}: $x_d = (u - c_u)/f_u$, $y_d = (v - c_v)/f_v$ gives $\mathbf{K}^{-1}_{2\times 2} = \mathrm{diag}(1/f_u, 1/f_v)$, and $\mathbf{J}_{D^{-1}} = \partial D^{-1}/\partial(x_d, y_d) \in \mathbb{R}^{2\times 2}$ is the Jacobian of the undistortion model (model-dependent; identity if no distortion). (x_n, y_n) do not depend on u_R.

Product rule. Since $p^c_x = Z \cdot x_n$, $p^c_y = Z \cdot y_n$, $p^c_z = Z$:

$$\frac{\partial(Z \cdot x_n)}{\partial(\cdot)} = \frac{\partial Z}{\partial(\cdot)}\, x_n + Z\, \frac{\partial x_n}{\partial(\cdot)} \tag{207}$$

Row 1: $p^c_x = Z\,x_n$.

$$\frac{\partial p^c_x}{\partial u_L} = Z\Big(C_{11} - \frac{x_n}{d}\Big) \tag{208}$$

$$\frac{\partial p^c_x}{\partial v_L} = Z\,C_{12} \tag{209}$$

$$\frac{\partial p^c_x}{\partial u_R} = \frac{Z\,x_n}{d} \tag{210}$$

Row 2: $p^c_y = Z\,y_n$.

$$\frac{\partial p^c_y}{\partial u_L} = Z\Big(C_{21} - \frac{y_n}{d}\Big) \tag{211}$$

$$\frac{\partial p^c_y}{\partial v_L} = Z\,C_{22} \tag{212}$$

$$\frac{\partial p^c_y}{\partial u_R} = \frac{Z\,y_n}{d} \tag{213}$$

Row 3: $p^c_z = Z$.

$$\frac{\partial p^c_z}{\partial u_L} = -\frac{Z}{d}, \quad \frac{\partial p^c_z}{\partial v_L} = 0, \quad \frac{\partial p^c_z}{\partial u_R} = \frac{Z}{d} \tag{214}$$

Assembled Jacobian.

$$\frac{\partial \mathbf{p}^c}{\partial(u_L, v_L, u_R)} = Z \begin{pmatrix} C_{11} - x_n/d & C_{12} & x_n/d \\ C_{21} - y_n/d & C_{22} & y_n/d \\ -1/d & 0 & 1/d \end{pmatrix} \tag{215}$$

Camera-frame covariance.

$$\boldsymbol{\Sigma}_{p^c} = \frac{\partial \mathbf{p}^c}{\partial(u_L, v_L, u_R)}\,\sigma^2_{\mathrm{px}}\,\mathbf{I}_3\,\frac{\partial \mathbf{p}^c}{\partial(u_L, v_L, u_R)}^{\top} \tag{216}$$

Body-frame covariance.

$$\boldsymbol{\Sigma}_{p^B} = \mathbf{R}_{Bc}\,\boldsymbol{\Sigma}_{p^c}\,\mathbf{R}_{Bc}^{\top} \tag{217}$$

Mono features. Initial covariance and point come from the point-velocity solver.

2.10 Search Region Construction

For filter feature points, the search region is the $n\sigma$ ellipse of the filter-predicted point with covariance $\mathbf{P}_k^-$. For all other points (interest and pre-admission features), the search region is constructed from the prediction function and the uncertainty in its inputs.

Prediction function. The predicted pixel position at frame k is a function of pose change, position, and velocity:

$$\hat{\mathbf{u}}_k(\Delta T,\, \mathbf{p}_i^{B_{k-1}},\, \mathbf{v}_i^{\mathbf{p}}) = \boldsymbol{\pi}\Big(\mathbf{R}_{cB}\,\Big(\Delta\mathbf{R}\,(\mathbf{p}_i^{B_{k-1}} + \mathbf{v}_i^{\mathbf{p}}\,\Delta t_{k-1}) + \Delta\mathbf{t}\Big) + \mathbf{t}_c\Big) \tag{218}$$

Uncertainty sets. The EKF-derived relative pose $\Delta\hat{T}$, $\boldsymbol{\Sigma}_{\Delta\xi}$ (Section 2.8.6) is always available. Position and velocity may or may not be available from previous iterations. When an estimate with covariance exists, the uncertainty set is the $n\sigma$ ellipsoid. When no estimate exists, the uncertainty set is a physically bounded domain.

Statistical sets ($n\sigma$ ellipsoids at confidence level α):

$$\mathcal{E}_{\boldsymbol{\xi}} = \Big\{\Delta\hat{T} \circ \mathrm{Exp}(\boldsymbol{\xi}) \;\Big|\; \boldsymbol{\xi}^\top\, \boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}\, \boldsymbol{\xi} \le \chi^2_{\alpha,6}\Big\} \tag{219}$$

$$\mathcal{E}_{\mathbf{p}} = \Big\{\mathbf{p} \;\Big|\; (\mathbf{p} - \hat{\mathbf{p}}_i)^\top\, \boldsymbol{\Sigma}_{pp}^{-1}\,(\mathbf{p} - \hat{\mathbf{p}}_i) \le \chi^2_{\alpha,3}\Big\} \tag{220}$$

$$\mathcal{E}_{\mathbf{p},\mathbf{v}} = \left\{\begin{pmatrix}\mathbf{p}\\ \mathbf{v}\end{pmatrix} \;\middle|\; \left(\begin{pmatrix}\mathbf{p}\\ \mathbf{v}\end{pmatrix} - \begin{pmatrix}\hat{\mathbf{p}}_i\\ \hat{\mathbf{v}}_i^{\mathbf{p}}\end{pmatrix}\right)^\top \boldsymbol{\Sigma}_i^{-1} \left(\begin{pmatrix}\mathbf{p}\\ \mathbf{v}\end{pmatrix} - \begin{pmatrix}\hat{\mathbf{p}}_i\\ \hat{\mathbf{v}}_i^{\mathbf{p}}\end{pmatrix}\right) \le \chi^2_{\alpha,6}\right\} \tag{221}$$

Bounded sets (physical limits, used when no estimate exists):

$$\mathcal{S}_Z = [Z_{\min},\; Z_{\max}] \qquad \text{Depth range} \tag{222}$$

$$\mathcal{S}_{\mathbf{v}} = \{\mathbf{v} \in \mathbb{R}^3 \mid \|\mathbf{v}\| \le v_{\max}\} \qquad \text{Velocity bound} \tag{223}$$

$$\mathcal{S}_{\mathbf{a}} = \{\mathbf{a} \in \mathbb{R}^3 \mid \|\mathbf{a}\| \le a_{\max}\} \qquad \text{Acceleration bound} \tag{224}$$

Acceleration expansion. For points with a velocity estimate, the velocity may have changed since the last frame:

$$\mathcal{E}_{\mathbf{p},\mathbf{v}} \oplus \mathcal{S}_{\mathbf{a}} = \{(\mathbf{p},\; \mathbf{v} + \mathbf{a}\,\Delta t_{k-1}) \mid (\mathbf{p},\mathbf{v}) \in \mathcal{E}_{\mathbf{p},\mathbf{v}},\; \mathbf{a} \in \mathcal{S}_{\mathbf{a}}\} \tag{225}$$

Search regions by available information. In each case, the search region is $\hat{\mathbf{u}}_k[\cdot] \cap [0,w] \times [0,h]$:

Available	**Input domain**
$\hat{\mathbf{p}}_i$, $\hat{\mathbf{v}}_i^{\mathbf{p}}$	$\hat{\mathbf{u}}_k[\mathcal{E}_{\boldsymbol{\xi}} \times (\mathcal{E}_{\mathbf{p},\mathbf{v}} \oplus \mathcal{S}_{\mathbf{a}})]$
$\hat{\mathbf{p}}_i$ only	$\hat{\mathbf{u}}_k[\mathcal{E}_{\boldsymbol{\xi}} \times \mathcal{E}_{\mathbf{p}} \times \mathcal{S}_{\mathbf{v}}]$
Neither	$\hat{\mathbf{u}}_k[\mathcal{E}_{\boldsymbol{\xi}} \times \mathcal{S}_Z \times \mathcal{S}_{\mathbf{v}}]$

Temporal matching protocol. The search region is a 2D subset of the pixel plane. Matching proceeds as follows:

1. *Discretise input domain.* Sample the uncertainty sets above. Each sample specifies a candidate (ΔT, $\mathbf{p}_i$, $\mathbf{v}_i^{\mathbf{p}}$).

2. *Project.* Pass each sample through $\hat{\mathbf{u}}_k$ (Equation 218) to obtain candidate pixels. The convex hull of these candidates, intersected with $[0, w] \times [0, h]$, is the search region. If intersect is empty, do not attempt match.

3. *Coarse match.* Evaluate SSD between the reference patch at $\mathbf{u}_i^{k-1}$ and the candidate pixels. Select the candidate with lowest SSD.

4. *Fine match.* Run pyramidal LK from the coarse winner to obtain sub-pixel accuracy.

5. *Validate.* Forward-backward check: run LK backward from the result; if $\|\mathbf{u}'' - \mathbf{u}_i^{k-1}\| > \epsilon_{\text{fb}}$, reject.

6. *Solver initialisation.* The input sample $(\Delta T,\ \mathbf{p}_i,\ \mathbf{v}_i^{\mathbf{P}})$ that produced the winning candidate provides the initial estimate $(\hat{\mathbf{p}}_i^0,\ \hat{\mathbf{v}}_i^{\mathbf{P},0})$ for the joint solver (Section 2.8.7). Point data is overwritten by stereo triangulation when available.

SSD rather than NCC is sufficient for temporal matching: the reference and target are from the same camera separated by one frame interval, so brightness and contrast are effectively unchanged.

2.11 Statistics

2.11.1 Chi-Squared Hypothesis Test

Following (Bar-Shalom et al., 2001) (p. 57), if $\mathbf{x} \sim \mathcal{N}(\bar{\mathbf{x}}, \mathbf{P})$ is an n-dimensional Gaussian, then:

$$q = (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{P}^{-1} (\mathbf{x} - \bar{\mathbf{x}}) \sim \chi^2_n, \tag{226}$$

with $\mathrm{E}[q] = n$ and $\mathrm{var}[q] = 2n$.

2.11.2 Feature Movement Detection

Individual gate. The EKF predicts measurement $\hat{\mathbf{z}}_{k,i} = h_i(\hat{\mathbf{x}}_k^-)$. The innovation:

$$\mathbf{y}_{k,i} = \mathbf{z}_{k,i} - h_i(\hat{\mathbf{x}}_k^-) \tag{227}$$

Under the stationarity hypothesis, $\mathbf{y}_{k,i} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_{k,i})$, where the innovation covariance combines state uncertainty projected into measurement space with sensor noise (Bar-Shalom et al., 2001) (p. 236):

$$\mathbf{S}_{k,i} = \mathbf{H}_{k,i}\, \mathbf{P}_k^-\, \mathbf{H}_{k,i}^\top + \mathbf{R}_{k,i} \tag{228}$$

The normalised innovation squared (NIS):

$$\gamma_{k,i} = \mathbf{y}_{k,i}^\top\, \mathbf{S}_{k,i}^{-1}\, \mathbf{y}_{k,i} \sim \chi^2_{\nu_i} \tag{229}$$

where $\nu_i = 4$ (stereo) or $\nu_i = 2$ (mono). If $\gamma_{k,i} > \chi^2_{\alpha,\,\nu_i}$, reject stationarity.

Joint consistency check. After individual gating, test the remaining inlier set $\mathcal{S}$ for collective consistency (Bar-Shalom et al., 2001) (p. 236):

$$\gamma_{\text{joint}} = \sum_{i\in\mathcal{S}} \gamma_{k,i} \sim \chi^2_{\nu_{\text{total}}}, \qquad \nu_{\text{total}} = \sum_{i\in\mathcal{S}} \nu_i \tag{230}$$

If $\gamma_{\text{joint}} > \chi^2_{\alpha,\,\nu_{\text{total}}}$: the inlier set is inconsistent.

2.11.3 Feature Admission

Before a feature point enters the EKF, it must pass a stationary test which depends on the correspondence type. For MS/SS/SM. correspondences:

$$\gamma_v = (\hat{\mathbf{v}}_i^{\mathbf{P}})^\top\, \boldsymbol{\Sigma}_{vv}^{-1}\, \hat{\mathbf{v}}_i^{\mathbf{P}} \sim \chi^2_3 \tag{231}$$

For MM correspondences:

$$\gamma_v = \frac{\hat{v}^2_{\perp,i}}{\sigma^2_{v_\perp}} \sim \chi^2_1 \tag{232}$$

If γ_v exceeds the threshold: do not admit. The NIS gate ejects features later if they were in fact moving. Points that pass are augmented into the EKF (Section 2.2.2) with covariance from the point-velocity solve $(A^\top A)^{-1}$

Limitation. For MM, motion along the epipolar plane is undetectable by this test (absorbed into the position estimate). The NIS gate (Section 2.11.2) catches such points one frame after EKF admission, since the epipolar geometry changes between frames.

3 Algorithm Overview

This section gives a conceptual map of the estimator. The code is the source of truth: algo.py contains the orchestration; tuning parameters are maintained in algorithm.yaml; Point-Set and `Point` datatypes are in points.py; the joint solver lives in solver.py; the EKF in ekf.py.

3.1 Point Sets and Roles

Tracked points are partitioned by role:

- $\mathcal{F}$: feature points augmented into the EKF state. Stationary, used as inertial references. Their measurement information enters via the pose update; they do not participate in the joint solver.
- $\mathcal{F}_{\text{pre}}$: candidate feature points awaiting their first two-frame solve and admission test. Enter the joint solver.
- $\mathcal{I}$: interest points steered by the focus. Not in the EKF. Enter the joint solver.

Each `Point` carries its pixel observations at $k-1$ and k (stereo or mono per camera), its 3D state and covariance in B_{k-1} from the previous solve and in B_k from the current solve, plus an id and age. The populated fields determine its stage (one-frame vs two-frame), correspondence type (SS/SM/MS/MM, see §2.8.7), and search strategy (§2.10).

3.2 Accumulator

The accumulator persists across iterations: the three point sets, the EKF posterior ($\hat{\mathcal{X}}^+_{k-1}$, $\mathbf{P}^+_{k-1}$), the previous image pair, the focus, and the last timestamp. At the end of each iteration k is pushed into $k-1$.

3.3 Input and Output

Input. Per frame: stereo pair (L_k, R_k), timestamp t_k, control input $\mathbf{u}_{k-1}$, focus point $\mathbf{F}_{k-1}$ and spread $\sigma_{F,k-1}$.

Output.

- Core EKF state $\hat{\mathcal{X}}^{\text{core}}_k = \langle T, \mathbf{v}, \boldsymbol{\omega}, \mathbf{g}^B, \mathbf{d}^B \rangle$ with $\mathbf{P}^{\text{core}}_k \in \mathbb{R}^{18\times 18}$. Feature rows and columns are stripped from the EKF state and covariance before output; internally the EKF retains them.

- Solver pose change $\Delta\hat{T}_{\text{solver}}$ and $\boldsymbol{\Sigma}_{\Delta\xi}$ from the joint solver. This is the recommended transform for propagating the downstream world model between frames.

- Point cloud $\mathcal{C}_k$ tagged by role and stage. Stage-2 points contribute $(\hat{\mathbf{p}}_i^{B_k}, \hat{\mathbf{v}}_i^{B_k}, \boldsymbol{\Sigma}_i^{B_k})$ from the joint solver; stage-1 (stereo-only) points contribute $(\hat{\mathbf{p}}_i, \boldsymbol{\Sigma}_{p,i})$ from stereo triangulation; feature points contribute position and covariance from the EKF state.

3.4 Pseudocode

Initialise ($k = 0$).

1: Receive (L_0, R_0, t_0, $\mathbf{F}_{-1}$, $\sigma_{F,-1}$)
2: FAST + Shi-Tomasi on L_0, R_0 (§2.7)
3: Focus-weighted grid selection into $\mathcal{I}$; grid selection into $\mathcal{F}_{\text{pre}}$ (§2.7)
4: NCC stereo match each point; stage-1 stereo triangulation for stereo pairs gives ($\hat{\mathbf{p}}_i$, $\boldsymbol{\Sigma}_{p,i}$) (§2.9)
5: Initialise EKF: ($\hat{\mathcal{X}}_0$, $\mathbf{P}_0$) with no feature points (§2.5)
6: Output ($\hat{\mathcal{X}}_0$, $\mathbf{P}_0$, $\mathcal{C}_0$); push accumulator $k \to k-1$

Iterate ($k = 1, 2, \ldots$).

— **Receive and propagate** —

1: Receive (L_k, R_k, t_k, $\mathbf{u}_{k-1}$, $\mathbf{F}_{k-1}$, $\sigma_{F,k-1}$); $\Delta t = t_k - t_{k-1}$
2: Propagate ($\hat{\mathcal{X}}_{k-1}^{+}$, $\mathbf{P}_{k-1}^{+}$) $\rightarrow$ ($\hat{\mathcal{X}}_{k}^{-}$, $\mathbf{P}_{k}^{-}$) (§2.2.1, §2.5)
3: FAST + Shi-Tomasi on L_k, R_k; store keypoint pool (§2.7)

— **Pre-update relative pose** —

4: Compute ($\Delta\hat{T}^{-}$, $\mathbf{\Sigma}_{\Delta\xi}^{-}$) from $\hat{T}_{k-1}^{+}$, $\hat{T}_{k}^{-}$ (§2.8.6)

— **Search and track** —

5: For each point with $\mathbf{u}_i^{k-1}$: build candidate set; SSD coarse match; pyramidal LK refinement; forward-backward validation (§2.10)
6: Track failures: drop from $\mathcal{I}$; marginalise from $\mathcal{F}$ (§2.2.2)
7: Stereo promotion: NCC search in other camera for mono points at k (§2.7)

— **EKF update** —

8: Per-feature NIS gate at $\chi^2_{\alpha_{\mathrm{NIS}},\,\nu_i}$; moving points demoted from $\mathcal{F}$ to $\mathcal{I}$ (§2.11.2)
9: Joint consistency check on surviving inliers; if fail: demote all $\mathcal{F}$ to $\mathcal{I}$ (§2.11.2)
10: Update on inliers plus gravity pseudo-measurement (§2.2.1, §2.6)

— **Post-update relative pose** —

11: Recompute ($\Delta\hat{T}^{+}$, $\mathbf{\Sigma}_{\Delta\xi}^{+}$) from $\hat{T}_{k-1}^{+}$, $\hat{T}_{k}^{+}$ — this is the pose prior for the joint solver (§2.8.6)

— **Joint solver** —

12: Stage-2 points in $\mathcal{F}_{\mathrm{pre}} \cup \mathcal{I}$:
13: Classify correspondence type (SS, SM, MS, MM); MM gated on $\|\mathbf{t}_{\mathrm{base}}\| > t_{\mathrm{base,min}}$
14: Build per-point position priors from stereo triangulation at $k-1$ and k where available (§2.8.7)
15: Solve jointly with pose prior ($\Delta\hat{T}^{+}$, $\mathbf{\Sigma}_{\Delta\xi}^{+}$) (§2.8.7)
16: Transform per-point ($\hat{\mathbf{p}}_i^{B_k}$, $\hat{\mathbf{v}}_i^{B_k}$, $\mathbf{\Sigma}_i^{B_k}$) via $\mathbf{T}_{\Delta,i}$

— **Feature admission** —

17: $\mathcal{F}_{\mathrm{pre}}$ points with first stage-2 solve: velocity consistency test (§2.11.3)
18: Pass: augment ($\hat{\mathbf{p}}_i^{B_k}$, $\mathbf{\Sigma}_{p,i}^{B_k}$) into EKF (§2.2.2)
19: Fail: demote to $\mathcal{I}$

— **Replenish** —

20: Retire points past their (per-instance jittered) $n_{\mathrm{max},i}$
21: Pre-emptive replenish: points reaching $n_{\mathrm{max},i}$ next frame get replacements now
22: Replenish $\mathcal{F}_{\mathrm{pre}}$ and $\mathcal{I}$ from keypoint pool; NCC stereo match; stage-1 stereo triangulation for new stereo points (§2.7, §2.9)

— **Output** —

23: Extract core EKF state $\hat{\mathcal{X}}_k^{\mathrm{core}}$, $\mathbf{P}_k^{\mathrm{core}}$ (strip feature rows/columns)

Assemble $\mathcal{C}_k$ from EKF (for $\mathcal{F}$), joint solver (for stage-2 $\mathcal{F}_{\text{pre}}$, $\mathcal{I}$), and stereo triangulation (for stage-1 $\mathcal{F}_{\text{pre}}$, $\mathcal{I}$)
return $\hat{\mathcal{X}}_k^{\text{core}}$, $\mathbf{P}_k^{\text{core}}$, $\Delta\hat{T}_{\text{solver}}$, $\boldsymbol{\Sigma}_{\Delta\xi}^{\text{solver}}$, $\mathcal{C}_k$
Push accumulator $k \rightarrow k{-}1$

4 Evaluation

4.1 Setup

We've implemented the algorithm in Python on the VID dataset (Zhang et al., 2022), using the sequence indoor_loadless_hovor_3096.1g_79.04s. The sequence is characterized by simple dynamics; no payload, no wind, and dominated by a hover-state. These characteristics make the sequence appropriate for first validation.

The dataset provides calibration, stereo video sequences and ground truth making it ideal for evaluating dynamics coupled VO systems.

Figure 1: Representative left-camera frame from the VID indoor sequence. Hangar interior at the start of the bag, drone stationary on the floor.

The sequence starts with the quadcopter stationary on the ground, then it takes off and climbs altitude, before it maintains hover, then finally it lands.

4.2 State Estimate Accuracy

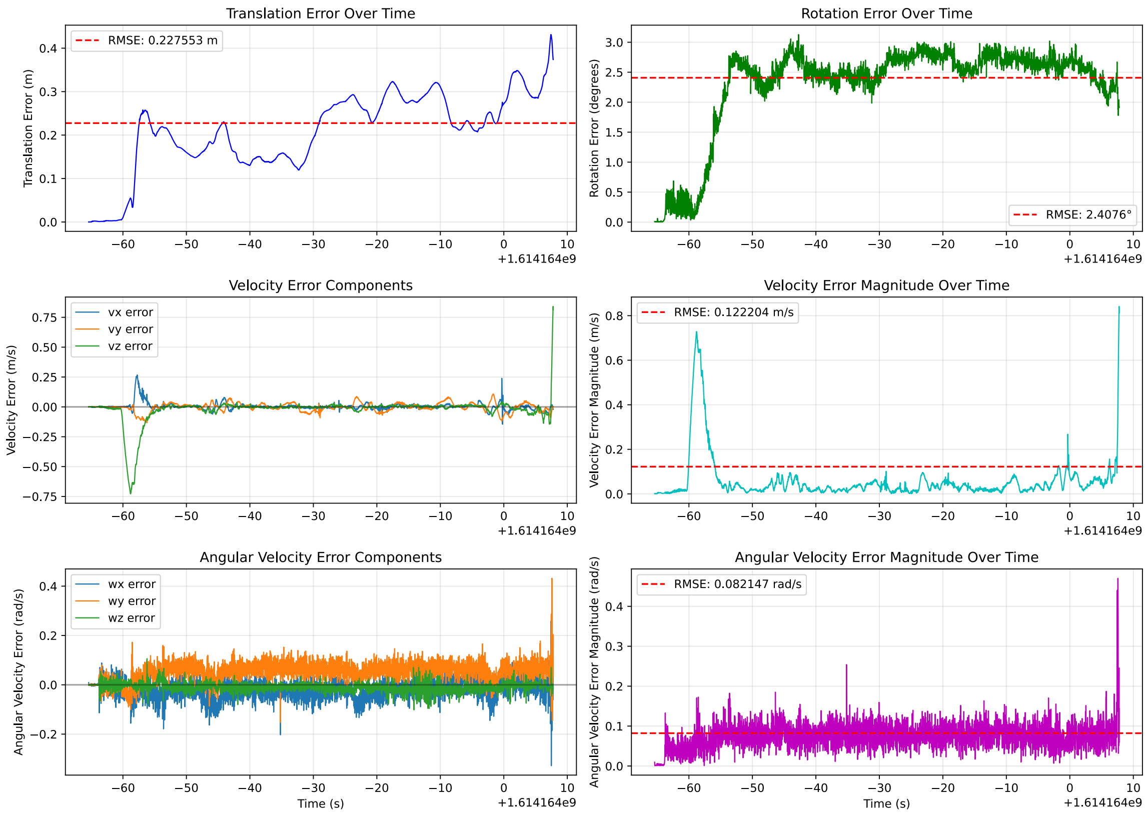

Figure 2: All but last 350 frames. Per-frame error magnitudes (translation, rotation, body-frame linear and angular velocity) against Vicon ground truth. Dashed red: RMSE over the full evaluated window.

Overall our state estimates appear accurate, but degrade during the transient off take-off and landing as illustrated by the large changes in error found at the beginning and end of the time interval. This is expected due to the discontinuous nature of the normal force, but we could likely alleviate the transient by increasing the process noise associated with the disturbance vector. Measurements observing the disturbance such as an accelerometer would also alleviate the transient.

We are pleased with the accuracy but recognize the need for tuning and optimization. For vision-only without IMU, this is reasonable first-implementation performance. Notably we observe non zero mean error in the pitch rate (wy error) and to a lesser extent in the roll rate (wx error) suggesting bias in our model which could stem from a non-zero displacement between the center of mass and geometric center.

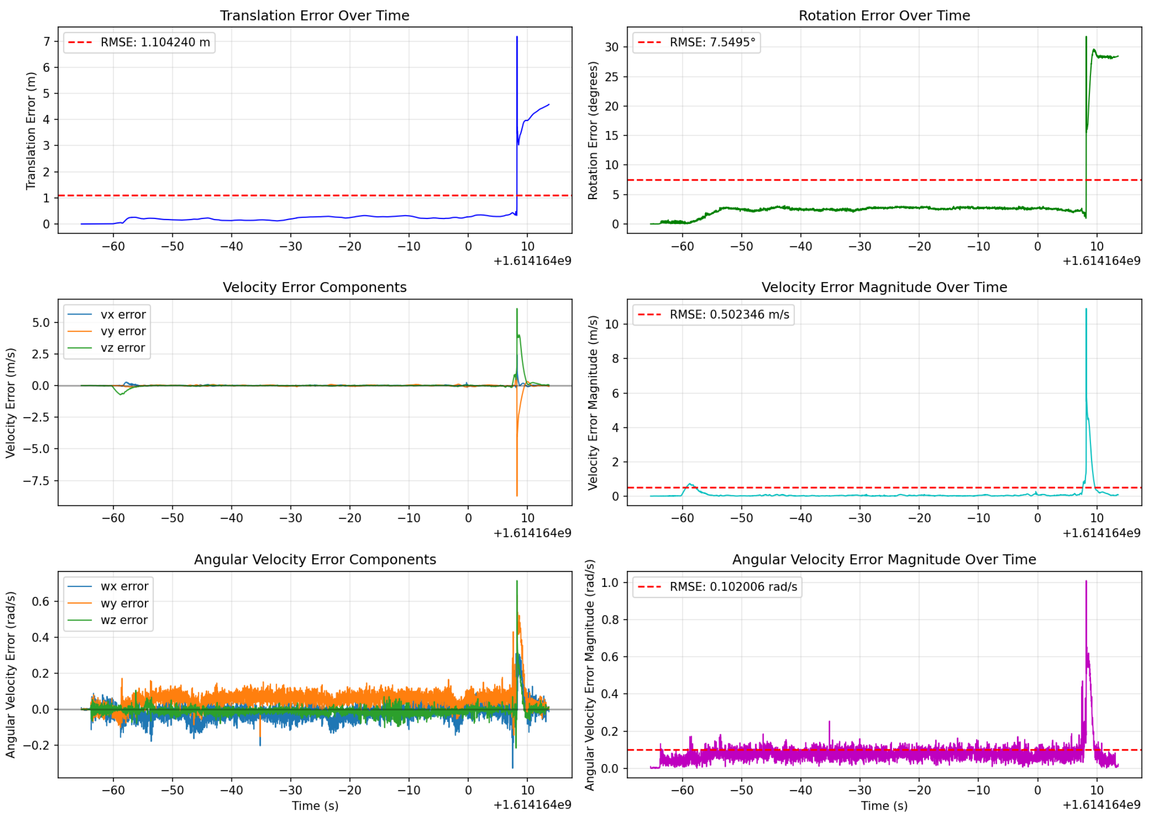

Figure 3: All frames. Per-frame error magnitudes (translation, rotation, body-frame linear and angular velocity) against Vicon ground truth. Dashed red: RMSE over the full evaluated window.

Above we see the error plots for the entire sequence. This illustrates the sharp increase in error which takes place during landing. Notably the system is able to recover with time.

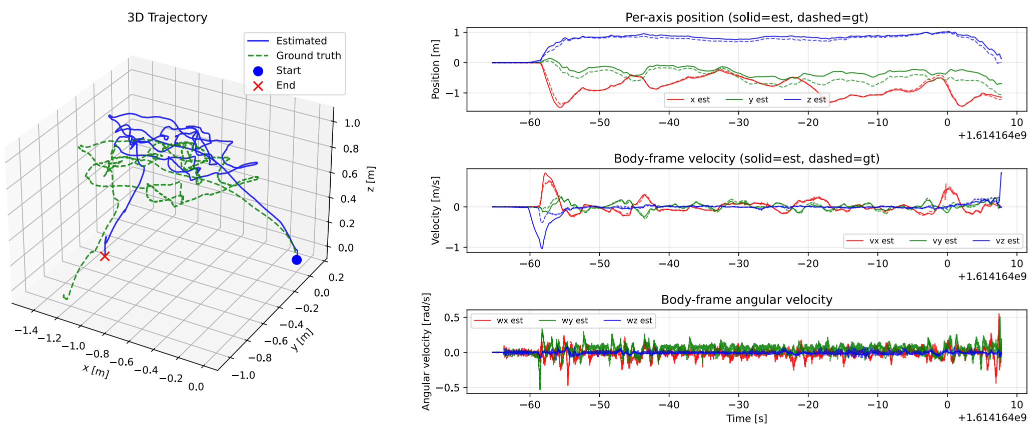

Figure 4: All but last 350 frames. Estimated vs ground-truth trajectory. Left: 3D path. Right: per-axis position, body-frame velocity, and angular velocity over time.

Quantity	**RMSE**	**Median**	**p95**
Translation [m]	1.104	0.229	3.964
Rotation [deg]	7.55	2.58	28.25
Velocity magnitude [m/s]	0.502	0.034	0.457
$\lvert v_x \rvert$ error [m/s]	0.065	0.010	0.065
$\lvert v_y \rvert$ error [m/s]	0.273	0.019	0.101
$\lvert v_z \rvert$ error [m/s]	0.417	0.010	0.420
Angular velocity magnitude [rad/s]	0.102	0.075	0.130
$\lvert \omega_x \rvert$ error [rad/s]	0.050	0.027	0.092
$\lvert \omega_y \rvert$ error [rad/s]	0.082	0.057	0.115
$\lvert \omega_z \rvert$ error [rad/s]	0.034	0.013	0.046

Table 1: All frames summary error metrics (4733 frames, 79.0 s), including ground contact, takeoff, flight, and landing. The RMSE figures are dominated by post-landing divergence.

The velocity numbers tell a clear story: median 1 cm/s, p95 (the 95th percentile) is 0.5 m/s. RMSE >> median means the error is highly concentrated in a small fraction of frames. That fraction is the takeoff and landing transient (see below).

Angular velocity is the opposite story: median≈RMSE means uniform noise floor, not transient-driven. The ω_y channel is the worst, this confirms the visible bias in the plots. ω_z is the best, ω_x middle. The asymmetry suggests among other: residual R_{BC} misalignment (camera-body extrinsic), vicon-marker-to-body alignment error, or an intrinsic limitation of vision-only rotation rate.

Translation median < RMSE << p95, suggests a small persistent offset (alignment or accumulated drift) and a transient spike.

Window [s]	**Rot [deg] med**	$\lvert v_z \rvert$ **med**	$\lvert v_z \rvert$ **p95**	**Trans med**
0–5	0.19	0.005	0.018	0.002
5–10	0.52	0.277	0.689	0.152
10–20	2.45	0.009	0.048	0.177
20–40	2.51	0.006	0.023	0.154
40–60	2.74	0.007	0.032	0.278
60–70	2.61	0.018	0.063	0.261
70–73	2.23	0.057	0.146	0.318
73–75	20.15	1.697	4.001	3.415
75–79	28.35	0.108	0.366	4.290

Table 2: Error breakdown by time window over all frames. Takeoff occurs at $t \approx 8$ s (window 5–10). Landing occurs at $t \approx 72$ s (window 70–73). The 73–75 s window contains catastrophic divergence: translation error grows from 0.32 m to 3.4 m and rotation from 2.2° to 20° within 2 s of touchdown.

4.3 State Diagnostics

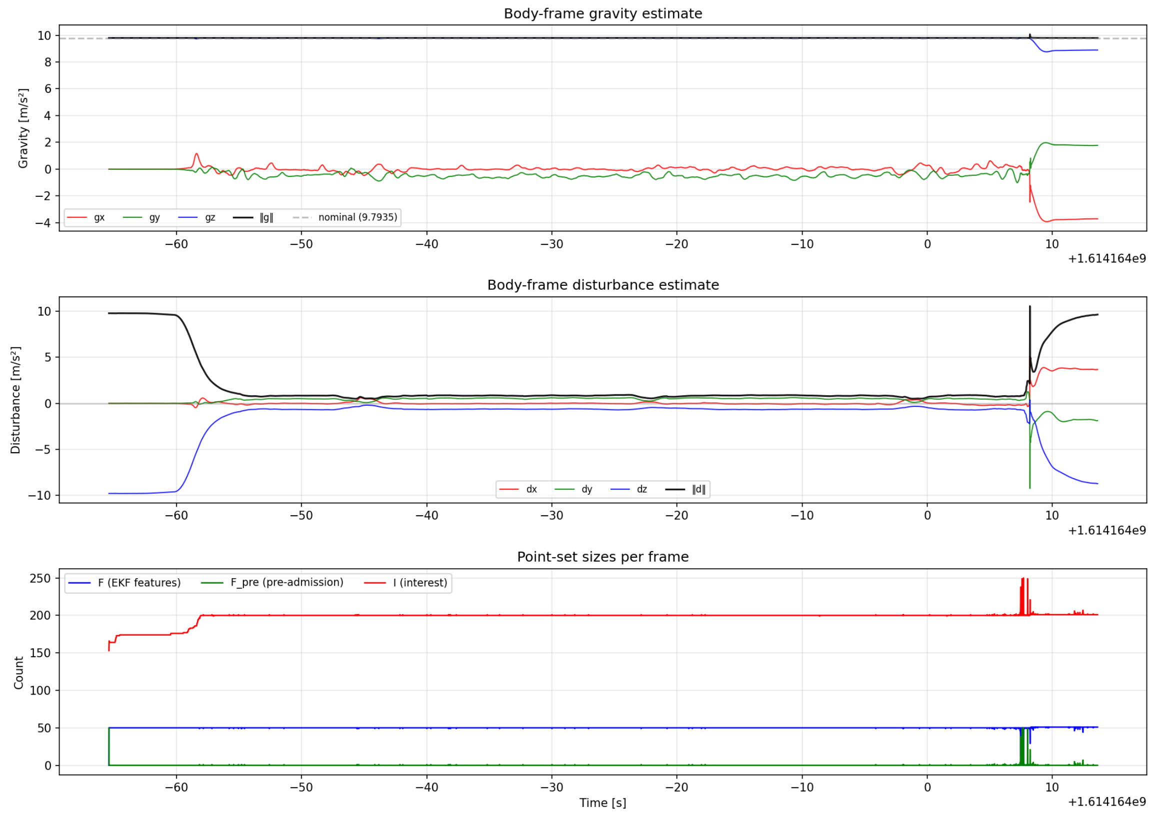

Figure 5: All frames. Internal state diagnostics. Top: body-frame gravity estimate (per-axis and magnitude; grey dashed: nominal $\|\mathbf{g}\| = 9.7935\,\mathrm{m/s^2}$). Middle: body-frame disturbance estimate. Bottom: point-set sizes over time — features in the EKF (blue), pre-admission candidates (green), interest points (red).

Gravity: $\|g\|$ stays locked at 9.79 throughout. The gravity pseudo-measurement is doing its job. Per-axis wandering (0.5 m/s^2) during the maneuver phase is gravity correctly being rotated through the body frame as the drone tilts.

Disturbance: the most diagnostic plot. The drone sits on the floor for $t < -55s$ with $d \approx (0, 0, -9.8)$. The normal force exactly cancels gravity. Then the drone lifts off, the filter correctly identifies that the normal force vanishes, d transitions to near zero. The transient takes 3-5 seconds, during which the filter is splitting the residual acceleration between gravity and disturbance.

This is exactly the takeoff discontinuity discussed in future-work, the EKF's Gaussian disturbance model cannot represent the instantaneous transition. The filter takes seconds to re-converge. With accelerometer integration this would resolve in one frame. Point counts: F holds at 50 (capped) once the cascade stabilises. I holds at 200 with small fluctuations from jittered n_{max} spreading marginalisations. $F_{pre} \approx 0$ steady state.

Points cycle through rapidly via admission or rejection. The smooth per-frame counts confirm the n_{max} jitter is working: no mass-marginalisation events visible except during landing. This is the NIS-gate firing off on high residuals between predicted and actual pixel observations. This is supposed to catch points with velocity, but here catches a degraded state which marginalises the current features points and selects new.

5 Future Work

This is a first draft. The work splits into three tiers: stabilise what exists/remove that which doesn't contribute above its alternative cost, extend the estimator, and build out the broader system. Items within a tier are ordered by priority. Tier 1 must precede Tier 2, adding capability to an unstable system multiplies failure modes without improving it.

5.1 Tier 1 — Stabilise

Fix what currently works fragilely.

Extend IterOutput to include $\mathrm{diag}(P_k^{\mathrm{core}})$ per frame. Required for NEES analysis, covariance consistency checks, and the closed-loop process-noise tuning sweep.

Joint solver.

- *Covariance propagation across frames.* The current implementation breaks the chain: previous-solve point posteriors are dropped and priors come from fresh stereo triangulation each frame. This loses information but is robust. The principled fix is full joint $\mathbf{\Sigma}$ propagation including pose-point cross-covariance, derived from the marginal of $(A^\top A)^{-1}$ transformed through the transport Jacobian (Eq. 199). Once correct, re-enable previous-solve priors.

- *MM rehabilitation.* Currently gated out entirely. The current gate is magnitude-only; it does not detect the degenerate case where $\mathbf{t}_{\mathrm{base}}$ is nearly parallel to the point ray. Investigate (a) direction-aware conditioning gate based on the effective Jacobian condition number, (b) Levenberg-Marquardt damping to control iterate divergence, (c) whether the issue is fundamental to 2-view monocular scene flow without a stereo prior or fixable in this architecture. Mature monocular scene-flow systems sidestep this via multi-frame parallax, photometric priors, or learned depth; we should understand which mitigation suits us.

- *Regularisation strategy.* Current stability relies on dual point priors (at $k-1$ and k), robustness through redundancy. Investigate whether Levenberg-Marquardt damping plus a single prior and a robust loss (Huber/Cauchy) gives equivalent stability without overcounting information.

- *Pose-prior approximation.* The Jacobian in A_0 uses identity, which is valid only near the prior mean. The exact form is $\mathbf{J}_l^{-1}(\bar{\boldsymbol{\delta}})$ from (Solà et al., 2021).

- *MM Jacobian completion.* The $\mathbf{d}_\perp$ derivative through ΔT is currently treated as constant per iteration; a full product rule should be derived and included.

- *Solver covariance growth.* Velocity uncertainty compounds across frames through the Δt-scaled coupling in $\mathbf{T}_{\Delta,i}$ (Eq. 197). Section 2.8.7 treats ΔT as exact during transport; including the pose-uncertainty contribution $\mathbf{J}_\xi \mathbf{\Sigma}_{\xi\xi} \mathbf{J}_\xi^\top$ corrects this.

- *Sparse normal equations.* The current solver inverts the dense $D \times D$ Hessian. The pose-points-cross structure admits a Schur complement: $O(N)$ instead of $O(N^3)$ on the point block. Necessary for real-time operation with $N > 100$ points.

EKF tuning and consistency.

- *Process noise calibration.* Current $\mathcal{Q}_u$, $\mathcal{Q}_d$, $\mathcal{Q}_\omega$ are priors, not validated against data. The filter is currently overconfident: the state collapses when measurements pause for ~ 10 frames. Implement an automated NEES sweep on a held-out flight, target NEES $\approx \nu$ for ν measurement dimensions.

- *Disturbance state behaviour.* Tight $\mathcal{Q}_d$ pins $\mathbf{d}^B$ near zero, preventing tracking of real disturbances. Either loosen $\mathcal{Q}_d$ or upgrade to a first-order Gauss-Markov model with appropriate correlation time.

- *Mass-marginalisation resilience.* Lifetime expiry currently causes coordinated extinction; the per-instance $n_{\max,i}$ jitter is a quick mitigation. Replace the randomised jitter with a deterministic admission-order stagger for reproducibility. Investigate rate-limited marginalisation (cap on features removed per frame).

- *Augmentation cross-covariance.* Currently uses a small seed ($\epsilon = 10^{-6}$) for the pose-feature cross-covariance at augmentation. The mathematically correct value is $\mathbf{J}_{\text{aug}} \mathbf{P} \mathbf{J}_{\text{aug}}^\top$ derived from the augmentation Jacobian. Replace the placeholder.

- *Relative-pose Schur consistency.* The post-update relative pose covariance occasionally produces negative eigenvalues on the order of -10^{-7}, significantly above floating-point noise floor. Currently masked by eigenvalue clipping at $\epsilon = 10^{-12}$. Likely source: the c_ids/b_ids selector construction across changing state dimension, or the $\mathbf{J}_r^{-1} \approx \mathbf{I}$ approximation. Trace and fix.

Calibration. Several parameters in calibration.yaml are flagged `[EST]`, educated guesses rather than measurements. Replace each with proper identification:

- Inertia tensor $\mathbf{J}$: bifilar pendulum measurement or MLE identification from flight data.

- Drag coefficient C_d: identify from steady-state hover or constant-velocity flight segments.

- Camera-to-body rotation $\mathbf{R}_{BC}$: currently the nominal mount; few-degree mounting misalignment is unmodelled. Run Kalibr against the dataset.

- Process and measurement noise parameters: see EKF tuning above.

Performance.

- *Vectorise the solver.* Current `build_normal_equations` issues $\sim 10^4$ JAX dispatches per frame through the per-point Python loop. Vectorise into batched NumPy: one matmul per camera per iteration instead of N small ones. Estimated 5–10× speedup. This is the prerequisite for the diagnostic loop that fixes everything else.

- *BLAS configuration.* Verify NumPy is linked against OpenBLAS or MKL rather than reference BLAS; set thread counts appropriately for the target machine.

5.2 Tier 2 — Extend

Add capability to the existing pipeline.

Solver-EKF integration.

- *State reformulation in B_k.* The current solver state lives in B_{k-1}; output requires transport to B_k. Reformulating the state directly in B_k would eliminate this step. Trade-off: pixel measurement at $k-1$ becomes non-linear in ΔT, but the output transport's covariance propagation is simplified.

- *Chained pose measurement.* The joint solver produces a refined $\Delta\hat{T}_{\text{solver}}$ that is currently not used by the EKF. Feeding it back as a pose-change measurement would close the feedback loop, but introduces information already used by both sides, requires careful tracking to avoid double-counting.

- *EKF features as solver point priors.* EKF feature points have position estimates with cross-covariance to the pose. They are currently excluded from the joint solver to avoid double-counting (their measurement information enters via the pose prior). With proper cross-covariance handling, including them as additional priors could improve solver conditioning.

Gyroscope, accelerometer, and RPM integration. Standard flight controller hardware (e.g. Betaflight-compatible boards with BMI270 or ICM-42688) packages a gyroscope and accelerometer on the same silicon die. Bidirectional DShot ESCs similarly provide rotor RPM telemetry for free. All three integrate into the existing EKF as sequential updates (§2.6) without changing the filter architecture.

Gyroscope. Provides angular velocity at 1–8 kHz. Measurement model: $\boldsymbol{\omega}_{\text{gyro}} = \boldsymbol{\omega} + \mathbf{b}_\omega + \mathbf{v}_\omega$, with gyroscope bias $\mathbf{b}_\omega \in \mathbb{R}^3$ added to the state as a slowly varying random walk. The bias is observable through the vision-derived angular velocity. Multi-rate updates between camera frames tighten the angular velocity estimate and improve propagation of all transport-dependent states.

Accelerometer. By the equivalence principle, a MEMS accelerometer measures specific force, the sum of all non-gravitational forces per unit mass. Gravity is absent from the reading because gravitational and inertial acceleration are locally indistinguishable: in freefall the proof mass and casing accelerate together and the sensor reads zero. The body-frame measurement model is therefore

$$\mathbf{a}_{\text{acc}} = \tfrac{1}{m}\,\mathbf{F}_{\text{thrust}}(\mathbf{u}) - \tfrac{C_d}{m}\,\mathbf{v} + \mathbf{d}^B + \mathbf{b}_a + \mathbf{v}_a \tag{233}$$

where $\mathbf{b}_a \in \mathbb{R}^3$ is accelerometer bias (random walk in the state) and $\mathbf{v}_a$ is white noise. No gravity subtraction is needed: $\mathbf{g}^B$ does not appear, unlike in standard VIO where recovering kinematic acceleration requires subtracting gravity, which requires knowing attitude, which couples to gyro bias. Our system estimates $\mathbf{g}^B$ directly from vision through the transport theorem, so the accelerometer is a clean, decoupled measurement. It directly constrains $\mathbf{v}$ (through drag), $\mathbf{d}^B$, and thrust-model accuracy, substantially improving disturbance observability, the accelerometer sees wind forces immediately, rather than waiting for them to propagate through feature point positions over multiple frames.

RPM telemetry. Bidirectional ESC telemetry provides measured rotor RPM, replacing commanded thrust $\mathbf{u}$ with actual thrust $T_i = k_f\,\omega_{r,i}^2$. This reduces effective actuator noise $\mathcal{Q}_u$, improving velocity and angular velocity propagation. The pose change covariance (§2.8.6) also benefits.

System model refinement and literature review. Survey existing literature on high-performance EKF-based quadrotor state estimation. A starting reference is (Svacha, 2019), which exploits aerodynamic effects with online parameter estimation. Aim to understand which model assumptions in our current system limit peak performance, and what mature systems do instead.
Concrete model upgrades, ordered by expected impact:

- Replace thrust command input with rotor RPM and introduce a thrust scale state.
- Optionally model rotor dynamics with first-order lag.
- Replace linear drag with anisotropic, velocity-dependent drag formulated on air-relative velocity.
- Avoid fixed aerodynamic coefficients; estimate online (thrust scale, drag coefficients, possibly inertia) where observable.
- Account for body–CoM and body–CoG offsets when these differ materially from the nominal alignment assumed in the dynamics (§2.4).
- Upgrade disturbance to first-order Gauss-Markov with both force and torque states.
- Account for thrust direction deviations due to inflow at high airspeed.
- Rebalance process noise to emphasise aerodynamic and disturbance uncertainty over actuator noise.

External measurements and discontinuity handling. Certain quantities should not be fed directly into the EKF but received from an external supervisory system that manages discontinuities and signal integrity.

Disturbances. The disturbance state $\mathbf{d}^B$ models smoothly varying external forces. Real disturbances can be discontinuous: the normal force vanishes at takeoff, wind disappears upon entering a building, collision forces appear instantaneously. With accelerometer integration many of these are observable directly, but the EKF's Gaussian model cannot represent a discontinuity. An external system can detect transitions and inject them as measurements: during stationary ground contact $\mathbf{d}^B = -\mathbf{g}^B$; a liftoff detector resets to zero via a disturbance pseudo-measurement.

GPS. A GPS receiver provides position in world frame W: $\tilde{\mathbf{t}}^W = \mathbf{t}^W_{B_t} + \mathbf{v}_{\text{GPS}}$. Integration requires the transform $T_{B_0,W}$ between the initial body frame and the GPS frame, estimable from the first few fixes. Since the body-frame formulation accumulates drift in T_{B_t,B_0}, GPS would periodically correct this. GPS signals are vulnerable to jamming and spoofing: signal-to-noise ratios and carrier-phase consistency detect jamming; change-point detection on the dead-reckoning vs. GPS residual detects spoofing (a spoofed signal deviates from the motion-model physics). The EKF should reject measurements that fail these checks, falling back to vision-only operation.

Passive/active IR sensor fusion. Replace one RGB camera with an IR camera or add a third camera. IR provides illumination-invariant features and object detection in low-visibility conditions. A pure RGB configuration is weak in poor lighting.

5.3 Tier 3 — Build

New capability beyond the estimator.

Dense RGB-DV output. Extend depth and velocity estimation toward dense output, with the focus directing computational resources. The joint solver's sparse seed points (position, velocity, covariance) anchor a local densification.

The scene as seen from B_t consists of independent surfaces; each pixel back-projects to a ray intersecting one. The sparse seeds sample these surfaces. From calculus, any C^1 surface is locally approximated by its tangent plane with error $O(r^2)$ in the distance r from the tangent point — valid over large regions for flat surfaces, only locally for complex geometry.

The densification algorithm starts at the densest region of the point cloud (point nearest the centroid) and diffuses outward (think along the lines of percolation or wave/heat propagation on a lattice). At each step: pick the current point and its two nearest neighbours, fit a plane (point + normal). For pixels near the plane centre, evaluate stereo photo-consistency (ZNCC) over a narrow depth range. Cleared pixels (passing consistency) expand the active region — diffuse outward in directions where pixels clear. When no more pixels clear for this plane, traverse the updated cloud, pick the next seed, fit a new plane, repeat. The process densifies the focus centre first and builds outward.

The clearing count self-regulates: many cleared pixels signal a large planar region (allow wider expansion); few signal curvature or a surface boundary (tighten or stop). Seeds with verified MAP depth disambiguate competing plane hypotheses. A fixed computational budget per frame ensures constant runtime regardless of scene complexity — complex scenes get less spatial coverage, simple scenes get more. Concepts from differential geometry and topology (local curvature estimation, surface connectivity, genus) may further inform plane selection, expansion criteria, and surface boundary detection.

Edge-bounded plane fitting. A closed surface in $\mathbb{R}^3$ projects to a 2D region in the image plane bounded by a closed curve (the silhouette/contour, the preimage of the Gauss map's critical set). Distinct surfaces in the scene project to distinct bounded regions, separated either by silhouette curves or by hard geometric edges (cube corners, roof ridges — C^0 but not C^1 boundaries).
Canny or similar edge detection extracts these boundary curves from each stereo image. The bounded regions they enclose correspond to single coherent surfaces. When fitting a plane to the sparse point cloud, restrict seed triplets to points whose projections fall within the same bounded region — these are guaranteed (or strongly likely) to lie on the same surface, and the plane normal from their triplet reflects the true local surface orientation.
This replaces the purely distance-based heuristic for seed selection with a topologically motivated one: points on the same projected region share a surface, regardless of their 3D distance. A C^1 surface within a region is well-approximated by its tangent plane in a neighbourhood. Hard edges (cube, fractal-like) are caught by the edge detector as C^0 boundaries and naturally partition the image into separate regions, each internally C^1. The stereo configuration provides cross-image verification: a region bounded by edges in the left image should have a corresponding region bounded by edges in the right image related by the plane's homography. This cross-verification rejects spurious edges (texture edges, shadows) from true silhouette/geometric edges.
The visual hull formalism could further exploit multi-view silhouettes for target model construction. Concepts from singularity theory (Whitney fold/cusp classification) and projective geometry (duality with tangent planes) may refine edge interpretation — e.g. distinguishing silhouette curves (where the surface turns away from the camera) from hard geometric edges (where the surface is non-smooth).

Internal world representation. The RGB-DV output provides per-frame point and velocity estimates with covariances, but a single frame may miss sub-pixel structures or suffer from occlusion and noise. A persistent internal representation aggregates information across frames: each RGB-DV output is projected into the representation and fused via a Bayesian update, so evidence accumulates over time. An object sub-pixel in one frame sweeps across multiple cells as the viewpoint changes; the probability it remains sub-pixel in all frames diminishes rapidly. The mathematical framework is well-developed: projecting measurements with known covariance into a volumetric or occupancy representation, updating cell beliefs, and maintaining the representation in a moving body frame. Frameworks to investigate: Bayesian occupancy grids, TSDFs, evidential grids, OctoMap, SLAM-style sparse maps.

Body-frame control. Design a control system that operates on the internal world representation, not on raw single-frame RGB-DV. The representation provides a temporally filtered, uncertainty-aware scene model from which planning and reactive avoidance can be derived. The body-frame formulation is a necessary foundation: conventional systems require world-frame state, which depends on external infrastructure (GPS, motion capture) unavailable in the target operating environment. Include simulation and real hardware testing.

Software and hardware optimisation. Implement the full pipeline in C++ with FPGA acceleration for the CV primitives (FAST, LK, NCC). Target: 60 Hz on low-cost edge hardware without GPU-class processors. The estimation mathematics (EKF, MAP solves) are negligible in cost; the bottleneck is image processing, which maps naturally to parallel hardware. For development, cgroup v2 (resource constraints comparable to edge hardware) and/or virtual hardware can stand in. The sparse normal-equations solver listed in Tier 1 is a prerequisite.
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